



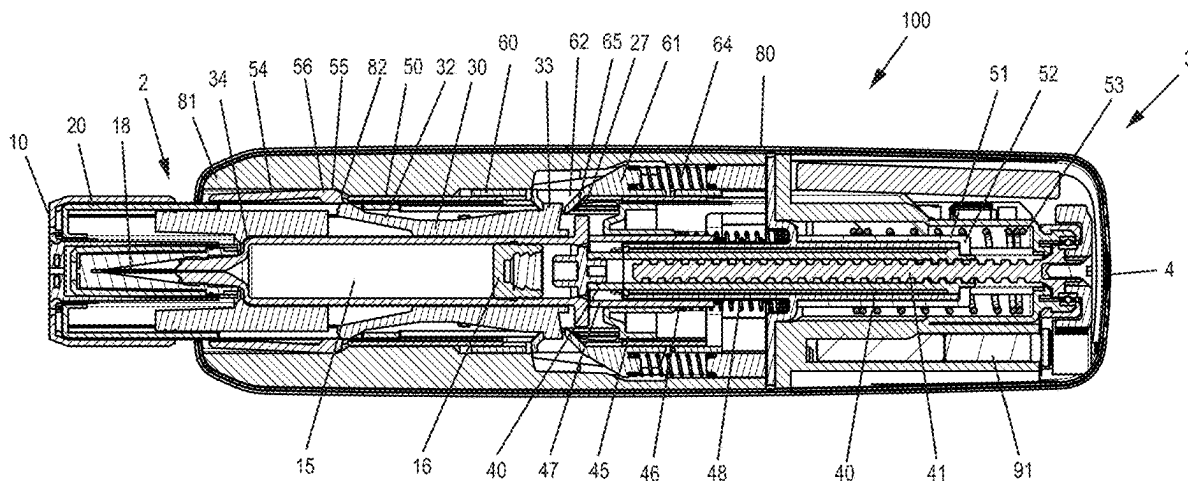
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(19) **United States**(12) **Patent Application Publication**
Scheurer et al.(10) **Pub. No.: US 2025/0276132 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **SAFETY MECHANISM FOR A DRUG
DELIVERY DEVICE****Publication Classification**(71) Applicant: **Ypsomed AG**, Burgdorf (CH)(72) Inventors: **Simon Scheurer**, Bern (CH); **Patrick
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2005/3263 (2013.01)(21) Appl. No.: **19/213,555**(22) Filed: **May 20, 2025****Related U.S. Application Data**(63) Continuation of application No. 19/033,106, filed on
Jan. 21, 2025, which is a continuation of application
No. PCT/EP2023/068250, filed on Jul. 3, 2023.(30) **Foreign Application Priority Data**

Jul. 22, 2022 (EP) 22186390.5

ABSTRACT

A safety mechanism for a drug delivery device for dispensing a liquid drug from a reservoir through a needle includes a housing defining a longitudinal axis, a needle cover movable along the longitudinal axis between a needle covering position and a retracted position, a movable lock element with a locking position to prevent the needle cover from being moved out of the needle covering position, and a trigger member movable along the longitudinal axis relative to the housing. The trigger member is operatively coupled to the lock element and a movement of the trigger member along the longitudinal axis causes the lock element to move out of the locking position allowing the needle cover to be moved out of the needle covering position.



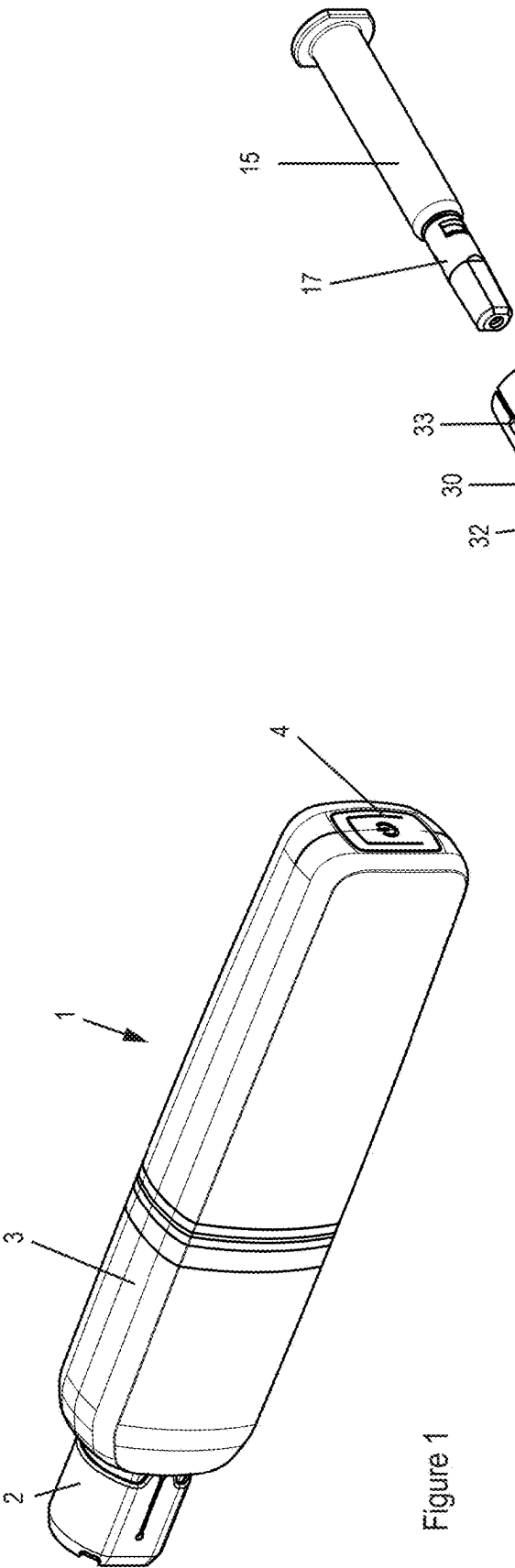


Figure 1

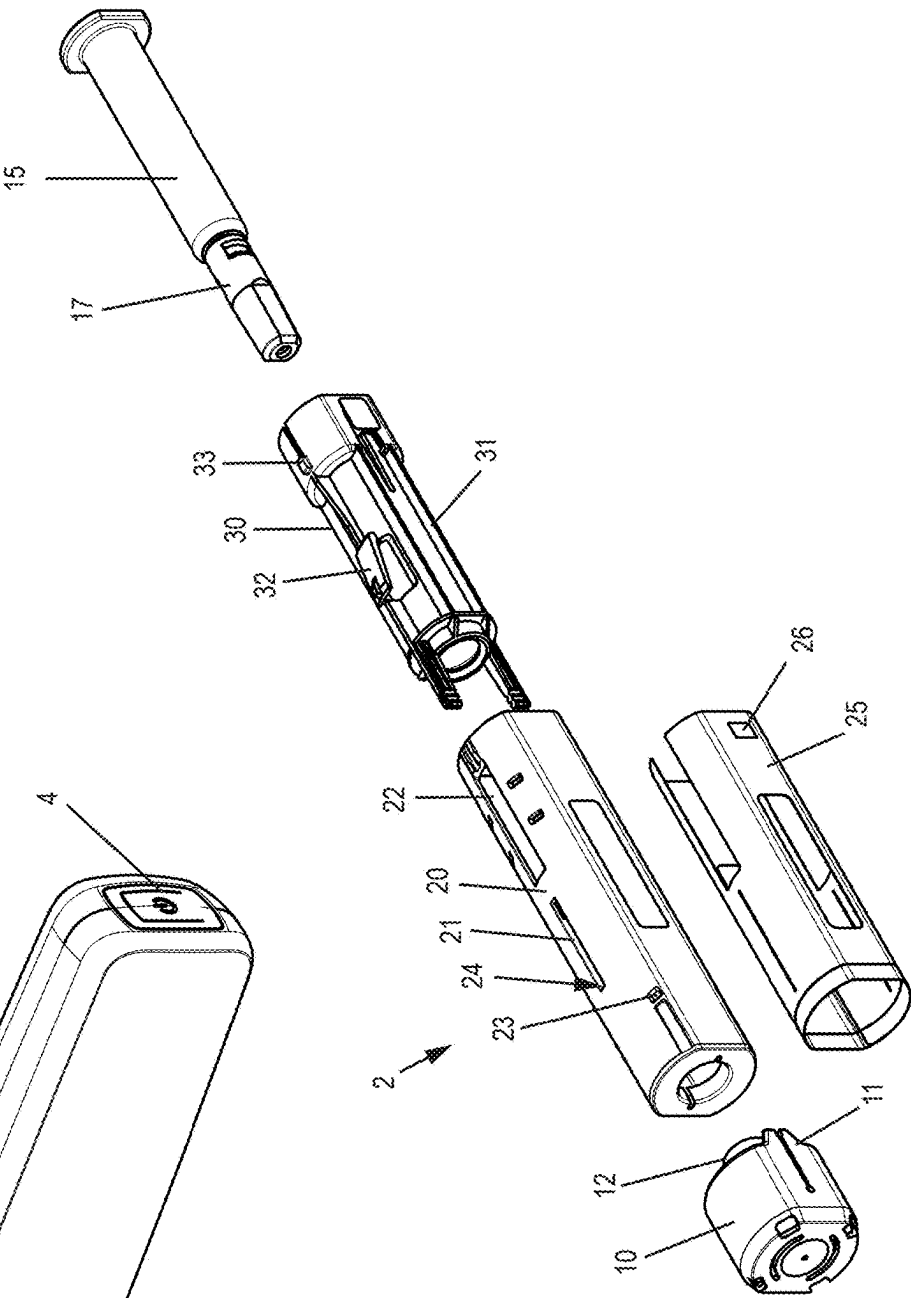


Figure 2

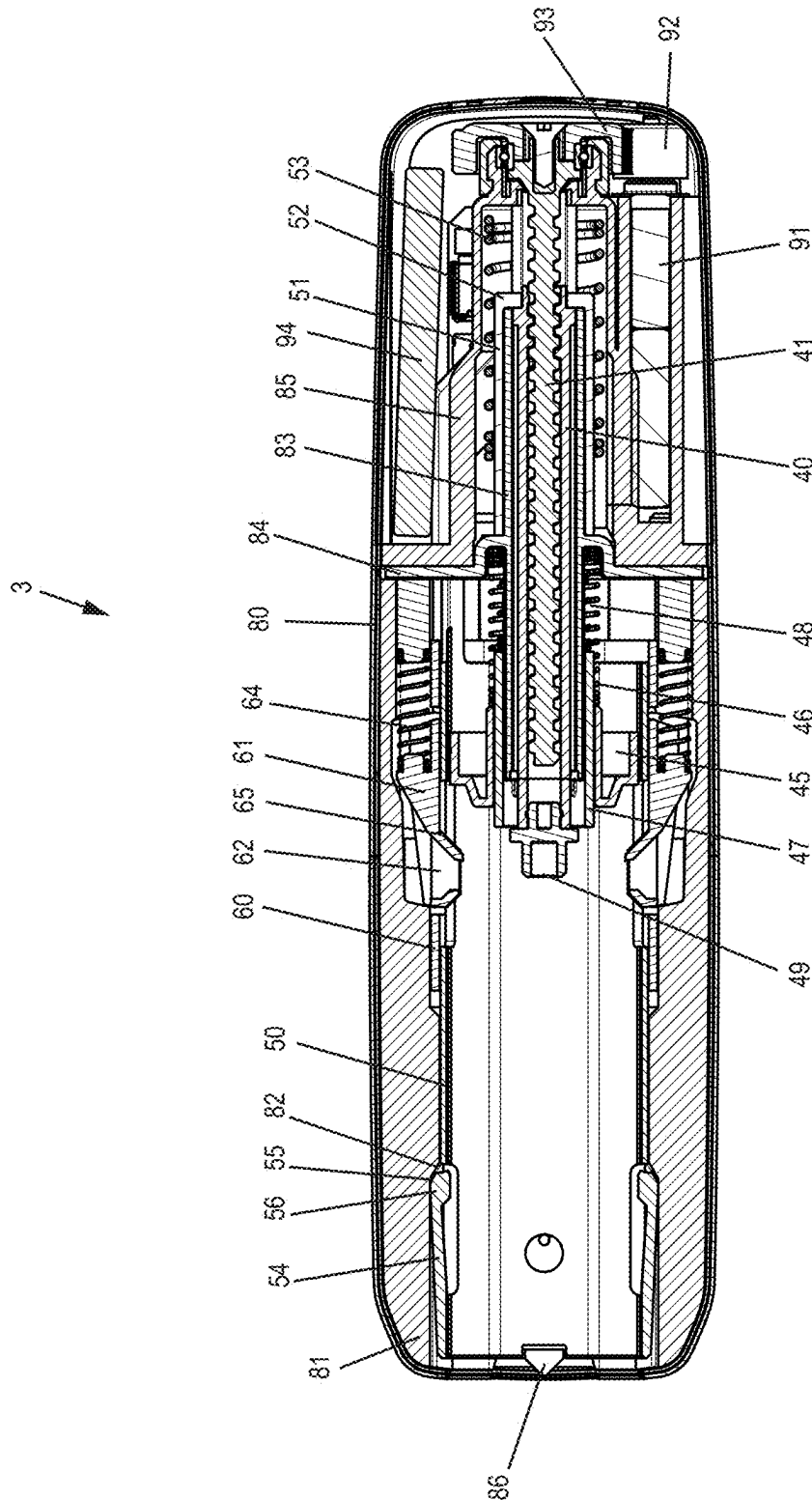


Fig. 3

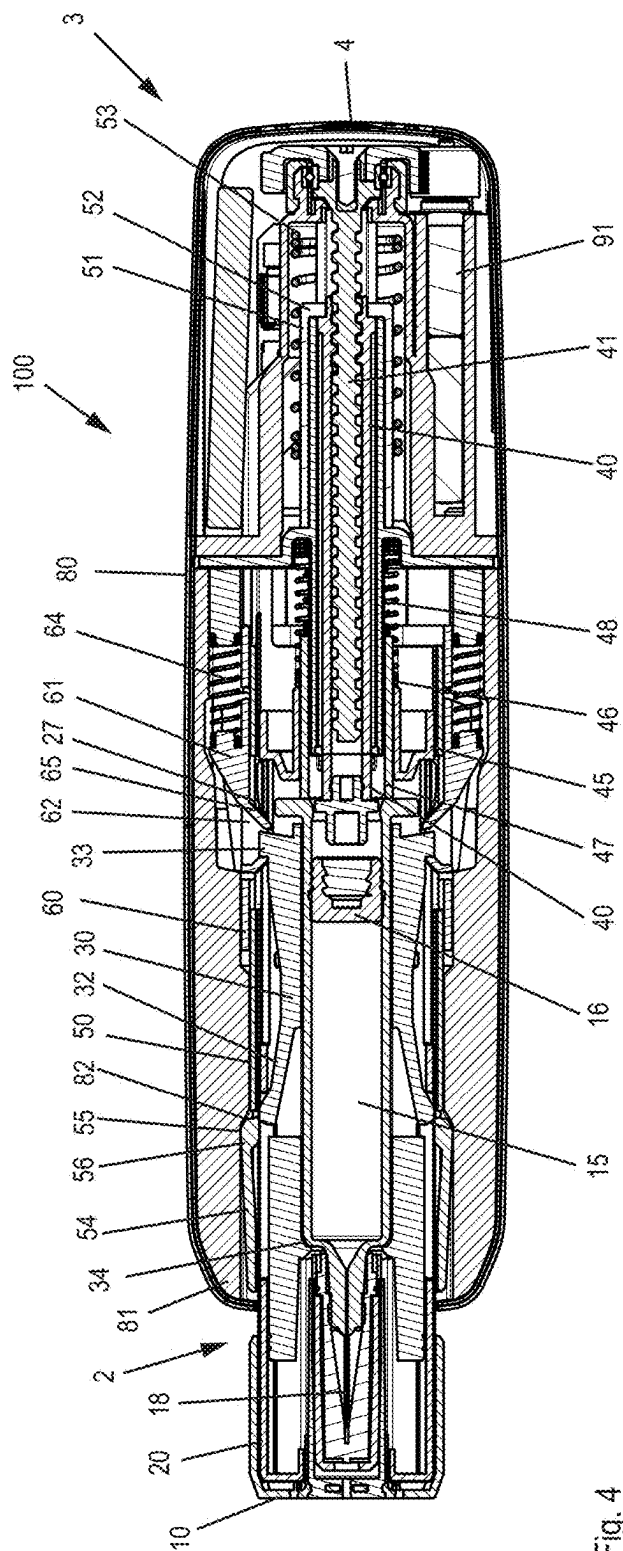


Fig. 4

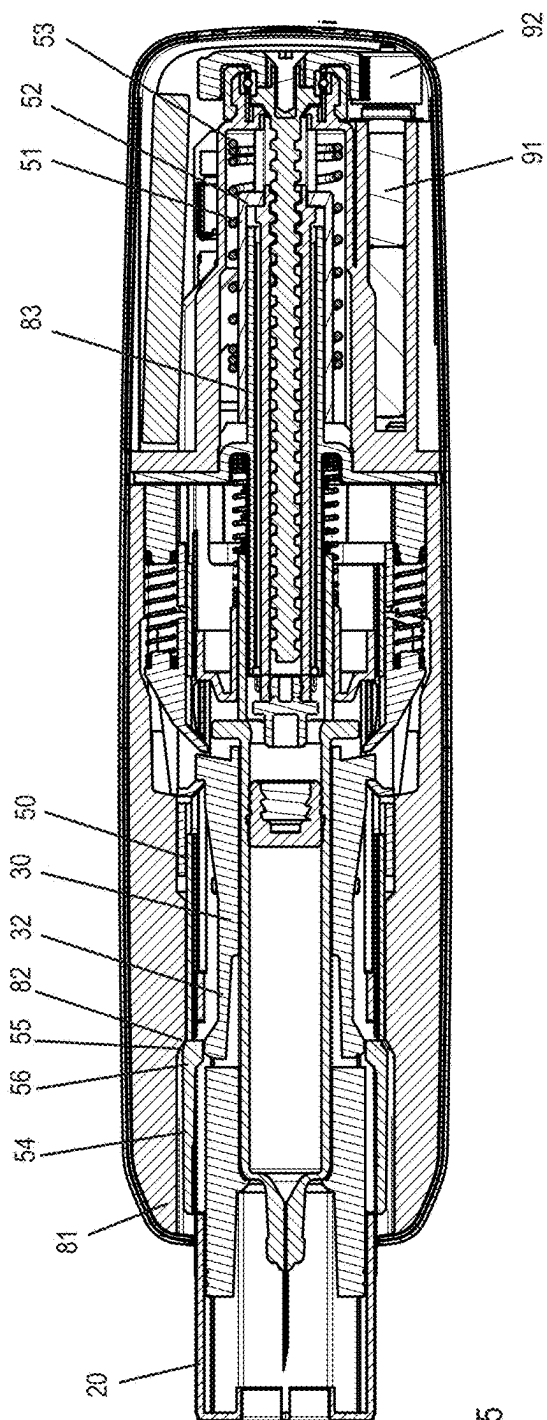


Fig. 5

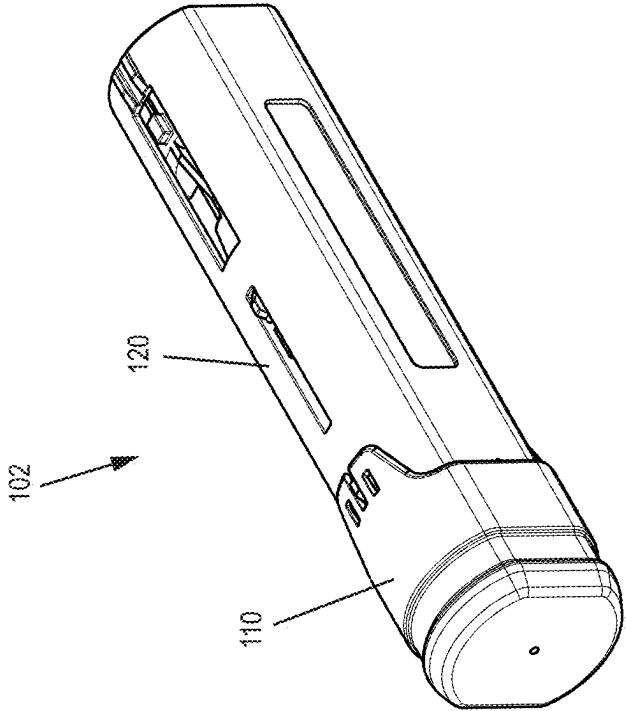


Fig. 6a

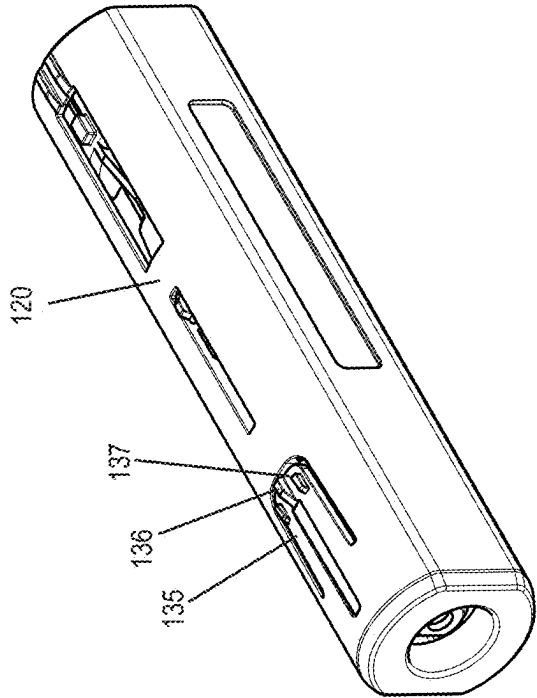


Fig. 6b

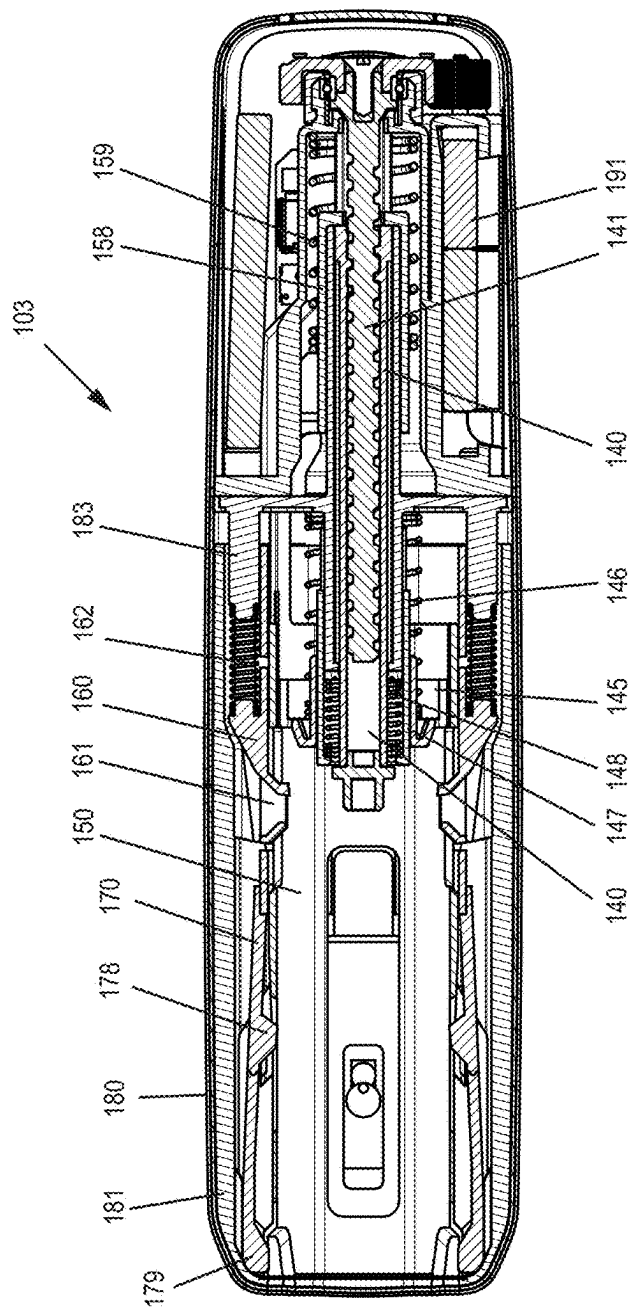


Fig. 7

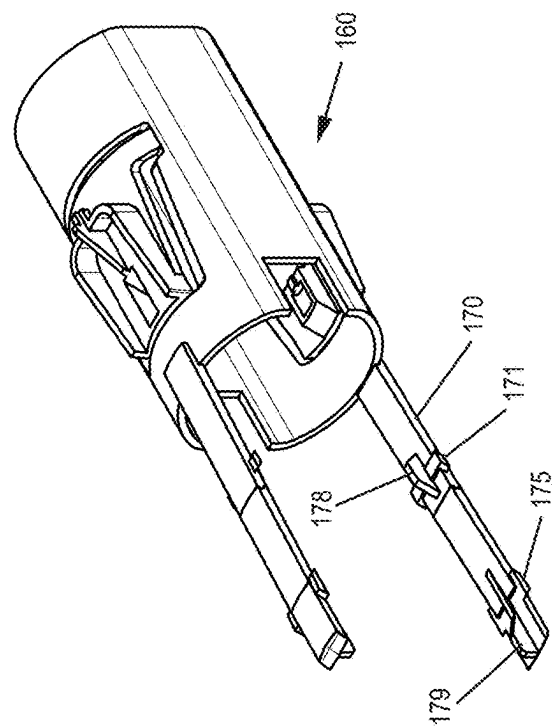


Fig. 8

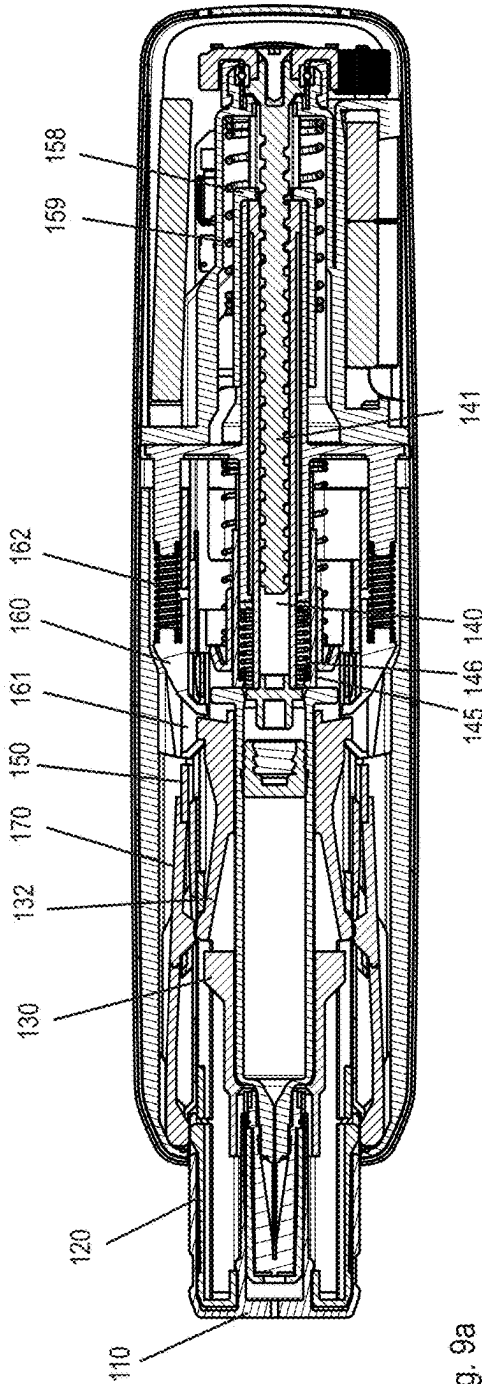


Fig. 9a

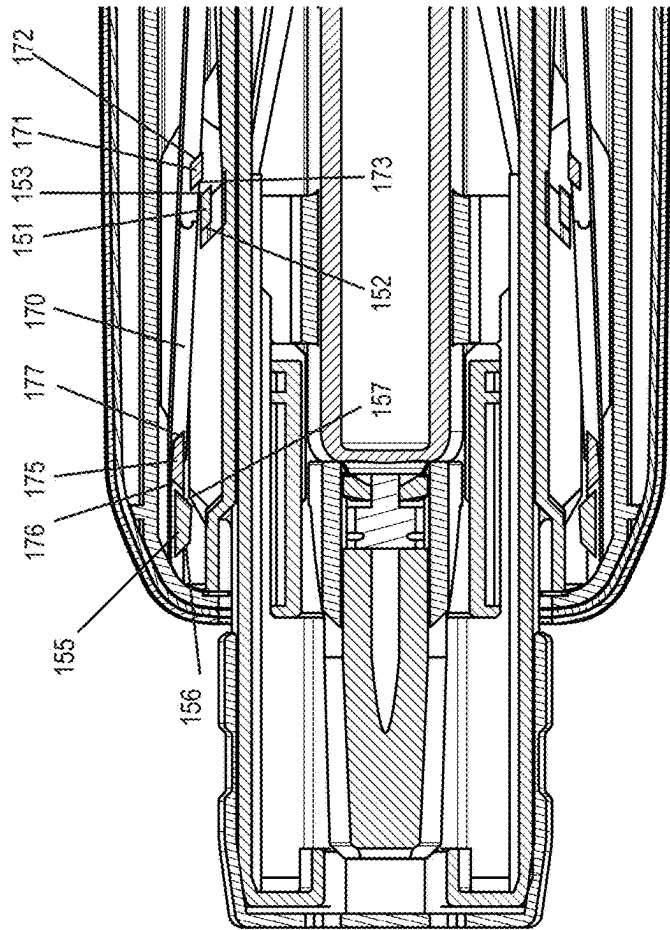


Fig. 9b

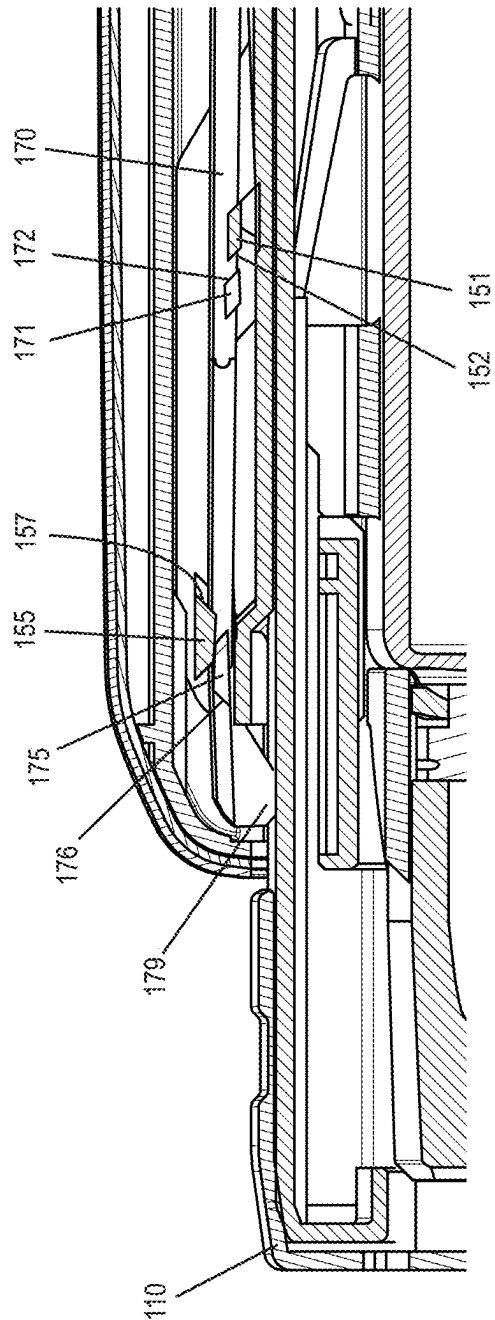


Fig. 10

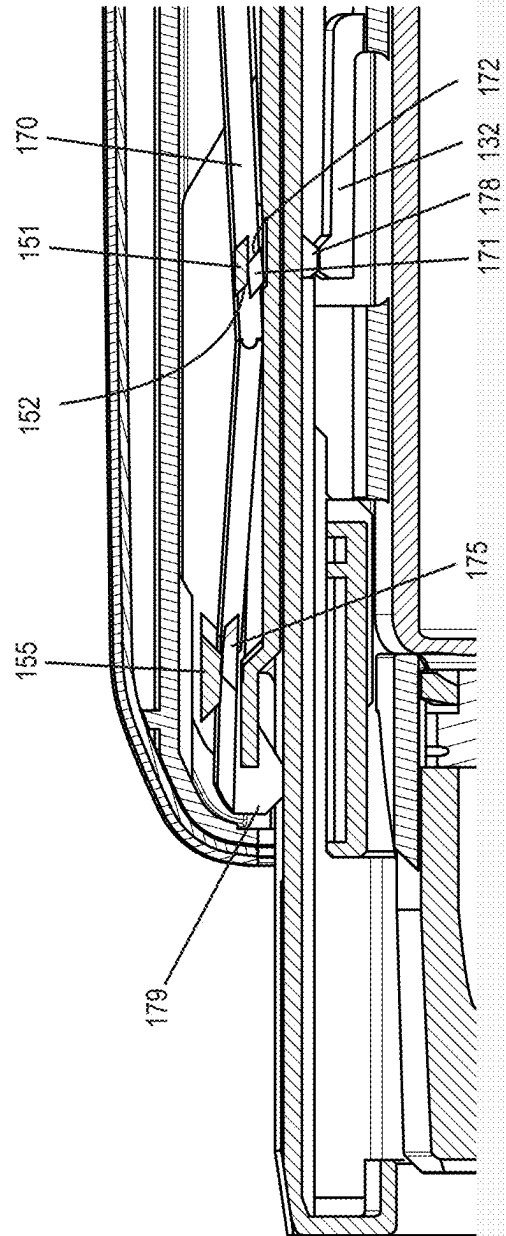


Fig. 11

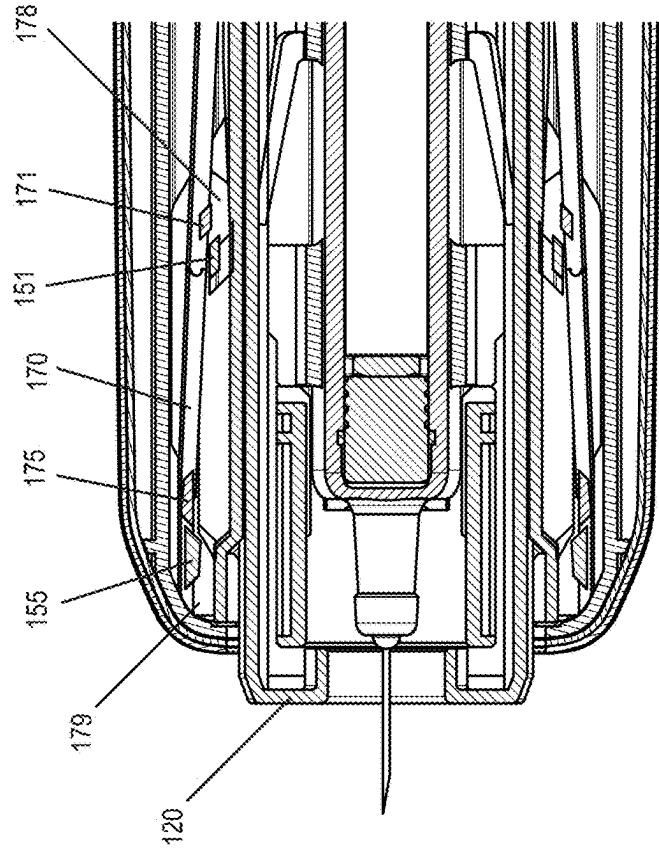


Fig. 12

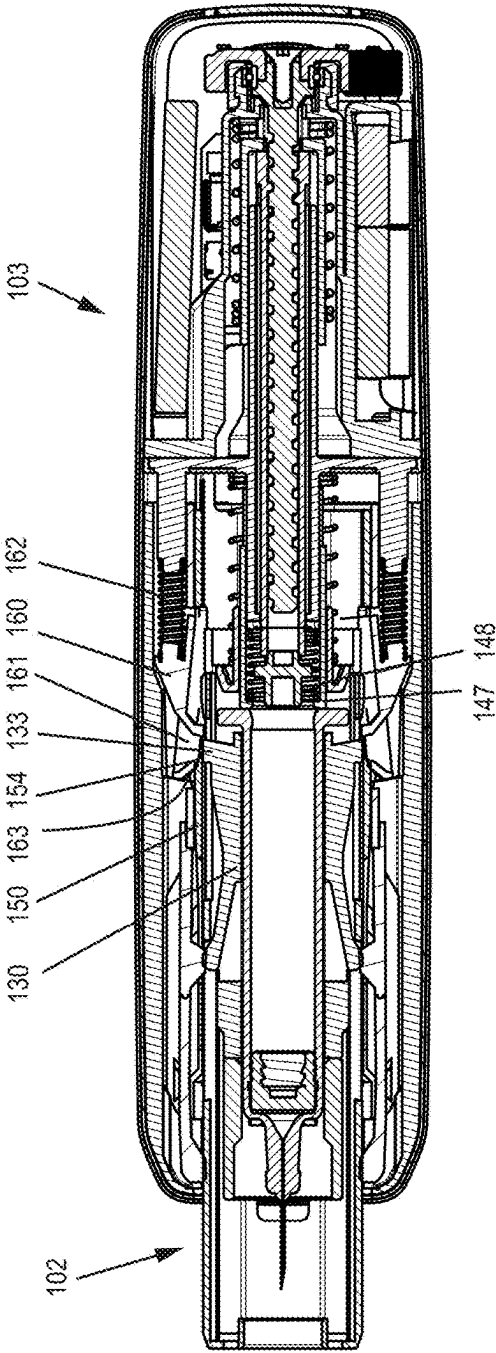


Fig. 13

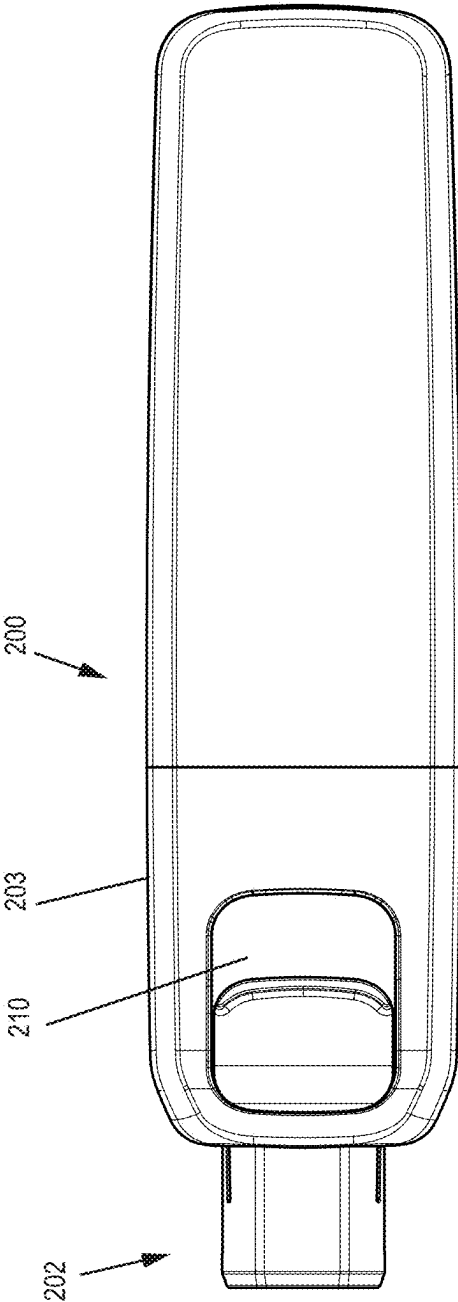


Fig. 14

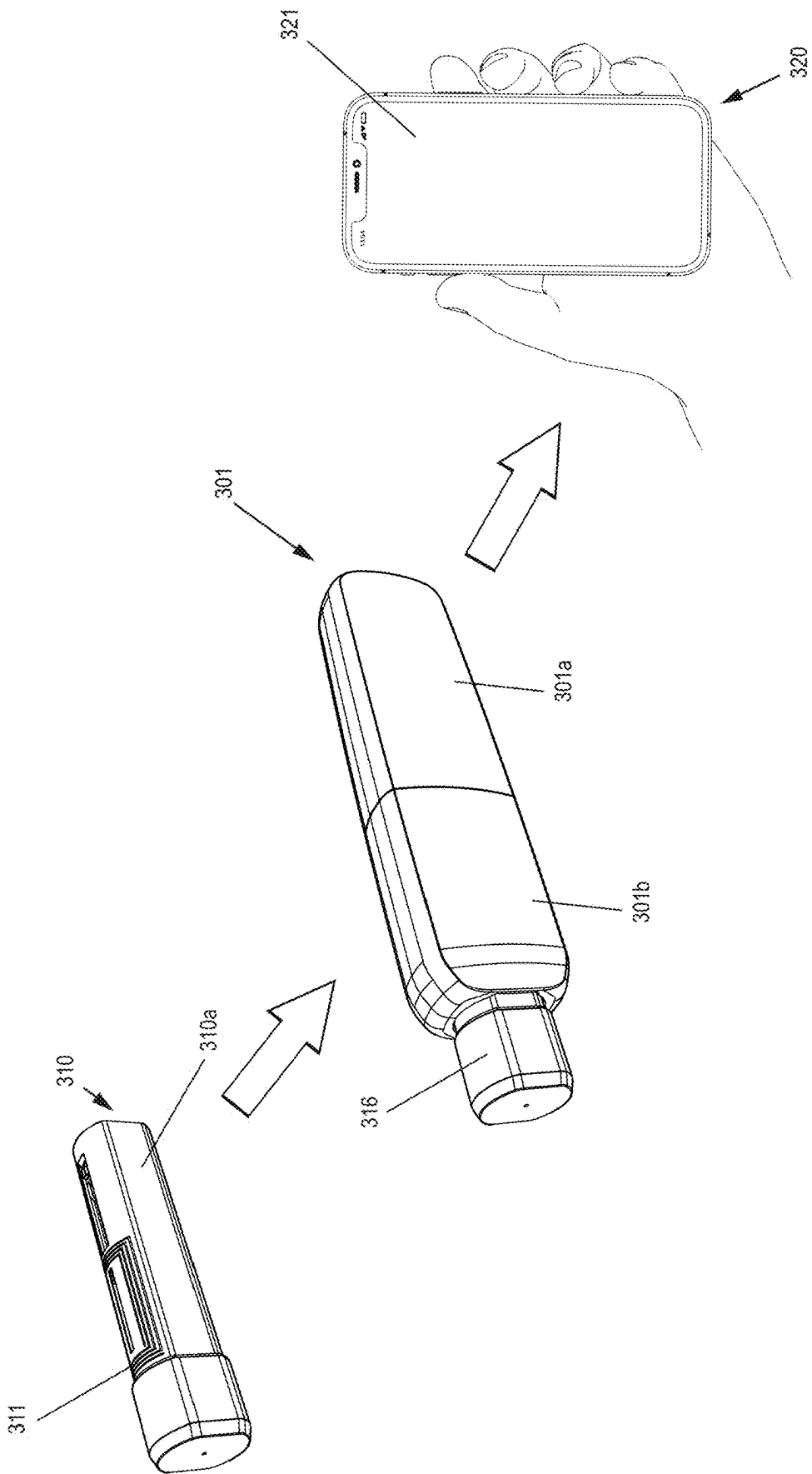
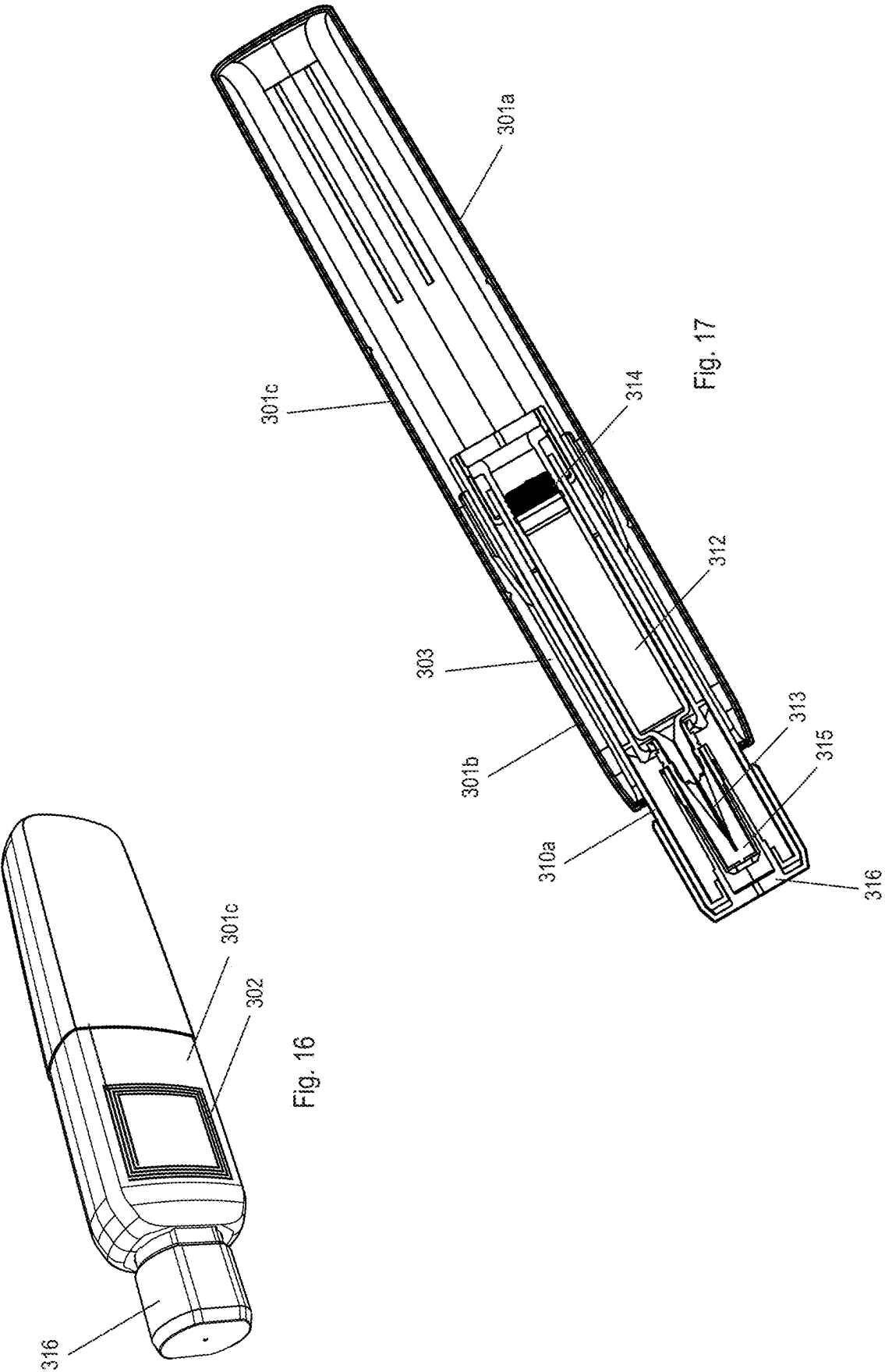


Fig. 15



SAFETY MECHANISM FOR A DRUG DELIVERY DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation of U.S. patent application Ser. No. 19/033,106, filed on Jan. 21, 2025, which in turn is a Continuation of International Patent Application No. PCT/EP2023/068250, filed Jul. 3, 2023, which in turn claims priority to European Patent Application No. 22186390.5, filed Jul. 22, 2022, each of which is incorporated by reference herein, in the entirety and for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to drug delivery devices or medicament delivery devices for injecting, delivering, administering, infusing or dispensing substances and/or liquids such as insulin, hormone preparations or vaccines. More particularly, a safety mechanism for a drug delivery device including a needle cover and a lock element that can be in a locking position, includes a lock element that engages the needle cover to prevent the needle cover from being moved out of a needle covering position.

BACKGROUND

[0003] A variety of diseases exist that require regular treatment by subcutaneous administration of a medicament, and a number of drug delivery devices have been developed to support a patient in accurately and controllably delivering an amount of drug in a self-administration process. Delivery devices include injection devices that are removed from the injection site after each medication event or drug delivery process, as well as infusion devices with a cannula or needle that remains in the skin of the patient for a prolonged period of time.

[0004] By way of example, diabetes may be treated by self-administration of insulin or its derivatives with the help of multi-variable-dose insulin injection pens. An injection pen device generally has an elongate device body defining a longitudinal main device axis. An automatic injection device has a motor or a drive spring for biasing a plunger rod and shifting a piston in a container barrel, wherein the drive spring may have to be charged or strained manually prior to injection of a dose. A manually powered delivery drive requires a user to manually provide the energy to move the piston, for instance by applying a distal force component to the injection device.

[0005] The medicament dose to be injected may be manually selected by turning a dosage knob and observing the actual dialed dose from a dose window or display of the insulin pen. A dose is dispensed by inserting the needle into a suited portion of human skin and by moving the piston manually or by pressing a release button of an automatic injection device. Automatic injection devices may include an electronic dose dial mechanism to automatically set a dose.

[0006] Also known is the use of autoinjectors with pre-filled syringes. Autoinjectors usually include a body for housing the syringe as well as an automatic drive mechanism in order to move the plunger of the syringe upon actuation of the autoinjector. The drive mechanism typically includes a source of drive, such as a motor or strong spring

for moving a transfer element, for example a rod, which acts on the plunger of the syringe.

[0007] For safety and hygiene reasons it is desirable that the needle does not protrude from a housing of the autoinjector with the exception of the time when the needle is used for injection of a medicament. Thus, either the autoinjector moves the needle out of the housing for the injection and back into the housing after injection or the housing provides a needle cover or needle guard which may be moved to unsheath the needle for injection and which may be moved back in a needle covering position after the injection.

[0008] The majority of autoinjectors are configured as single use devices which incorporate both the syringe and the drive mechanism in the same housing. Such devices are usually disposable for hygiene reasons.

[0009] Disposable autoinjectors including an electric actuator or an electronic control require a source of energy which is usually in the form of a battery. However, in this case, the autoinjectors should not be disposed in the regular waste, but have to be subjected to special disposal or to recycling, which is an additional burden to the patient. Further, disposing a battery, motor and/or electronics after a single use is a waste of resources and increases the costs of the autoinjector.

[0010] In order to account for a need to handle and dispose the autoinjector parts differently semi-reusable autoinjectors have been developed. Such autoinjectors typically include a reusable drive unit as well as a disposable syringe unit which may be releasably coupled or attached to the drive unit. The drive unit usually includes the drive mechanism and electronics whereas the syringe unit includes the syringe with the injection needle and a needle cover sleeve. The user can thus dispose of the syringe unit when empty or after use and can load the drive unit with a new syringe unit for an upcoming injection.

[0011] Different mechanism for covering and uncovering the injection needle are known in the art. Typically, a cover sleeve or movable sheath can be unlocked and subsequently retracted to expose the needle and prepare the delivery device for an injection.

[0012] WO12038721 A1 discloses a semi-reusable autoinjector with a syringe unit including a syringe and with a drive unit including a torsion spring and a gear to automatically dispense the drug from the syringe. The drive unit includes an opening to accommodate the syringe unit. In a bottom plate in the opening two forwardly projecting lugs are designed to enter into recesses in the syringe unit housing upon insertion of the syringe unit to unlatch a rearward collar of a syringe shield. This allows the syringe shield to move to a retracted position.

[0013] WO 2010/046569 A1 discloses a reusable autoinjector with a syringe unit that can be coupled to a drive unit by a bayonet connection. Once a cam of the drive unit is inserted in a slot in the syringe unit housing a cover sleeve is unlocked and can be moved to unsheath the needle.

[0014] WO 2021/069106 A1 discloses an autoinjector including a syringe unit and a drive unit. When an axially protruding locking pin of the drive unit is pushed through in a pin opening in the syringe unit, it comes in contact with a syringe holder arm. This results in a deflection of the holder arm, which in turn unlocks a needle cover sleeve and allows it to move in the proximal direction.

SUMMARY

[0015] It is an objective of the disclosure to provide a space-saving design and a safe and reliable release of a needle protection of a drug delivery device.

[0016] According to the disclosure a safety mechanism for a drug delivery device for dispensing a liquid drug from a reservoir through a needle is provided. The safety mechanism includes:

[0017] a housing defining a longitudinal axis, the housing may be adapted to accommodate a dispensing mechanism for dispensing the drug and is adapted to accommodate at least partially the reservoir with the drug;

[0018] a needle cover movable along the longitudinal axis between a needle covering position, in which the needle is covered by the needle cover and a retracted position, in which the needle cover is moved away from a tip of the needle and the needle is exposed and protrudes from the housing;

[0019] a movable lock element that can be in a locking position, in which the lock element may engage the needle cover or a needle cover supporting member or the housing to prevent or block the needle cover from being moved out of the needle covering position towards a retracted position.

[0020] The safety mechanism further includes a trigger member movable along the longitudinal axis relative to the housing. The trigger member may be movable exclusively along the longitudinal axis. The trigger member is operatively and permanently coupled (direct or indirect via an intermediate member) to the lock element and can actuate the lock element such that a movement of the trigger member along the longitudinal axis causes the lock element to move out of the locking position. Thus, the needle cover is no longer blocked by the lock element and the needle cover can be moved out of the needle covering position towards a retracted position. The delivery device is then ready for an injection as the needle is exposed and accessible. In other words, with an exclusive longitudinal movement of the trigger member (manually or automatically driven) the needle cover can be unlocked and retracted towards an uncovering position.

[0021] In contrast to known approaches that focus on a radial release trigger (in a plane perpendicular to the longitudinal axis), according to the disclosure, the trigger member is movable along the longitudinal axis. As the trigger member does not move in the radial direction the dimensions of the safety mechanism and thus the dimension of the drug delivery device in the radial direction can be reduced compared to known mechanism. Thus, the arrangement according to the disclosure provides for a space-saving design. In view of WO12038721, the disclosure provides at least an alternative design with a trigger member movable relative to the housing.

[0022] The trigger member unlocks the locking element and thus the trigger member may also be referred to as an unlocking member. The trigger member may be, for example, a sleeve or a nut movable relative to the housing. The trigger member may be moved manually by a user force or automatically by an automatic drive such as an electric motor or a preloaded spring member. The trigger member may be arranged inside the housing of the safety mechanism or inside a housing which encloses the safety mechanism and a dose and dispensing mechanism. The trigger member

is thus not arranged in an injection device cap or in an external device. The trigger member may be sleeve-shaped and coaxially arranged to a reservoir or syringe inside the housing. This allows for a space-saving design.

[0023] Furthermore, the trigger member can actuate the lock element. That means the lock element can be brought into contact with the lock element either directly or indirectly via an intermediate member. In the latter case the trigger member contacts or engages the intermediate member which in turn contacts or engages the lock element. In this way, a movement of the trigger member is transferred to the lock element to move the lock element out of the locking position towards a release position in which the lock element does not hinder or block the needle cover and the needle cover is movable out of the covering position towards a retracted position.

[0024] Optionally, the longitudinal shifting or movement of the trigger member can be transferred to a radial deflection or movement of the lock element, either directly or indirectly via an intermediate member.

[0025] The lock element is adapted to block or hold the needle cover in the needle covering position. The lock element may be a deflectable, rotatable or shiftable element, for example in form of a ledge, arm, protrusion or knob. Further, the lock element may be a movable element of the needle cover or of a housing part adapted to contact, abut or engage the needle cover.

[0026] The lock element can be in the locking position and may be moved into an unlocking position, intermediate position or end position. If the lock element is in the locking position it may abut a stop surface of the needle cover or of a support or housing element to prevent the needle cover from being moved out of the covering position. If the lock element is not in the locking position it does not prevent the needle cover to be moved out of the covering position. The lock element may be part of the needle cover, of a support element for the needle cover of the housing or any other part of the drug delivery device.

[0027] The needle cover in turn is movable along the longitudinal axis and may be guided within the housing or a member connected to the housing or a member of a reservoir unit attachable to the housing. The movement of the needle cover allows to switch between a safety state in which the needle is covered and the needle does not protrude from the housing and an injection state in which the needle is uncovered and exposes. The needle cover may be implemented as shield or sleeve or sleeve-shaped element adapted to cover or envelop the needle of the drug delivery device the covering position.

[0028] The drug delivery device may be an injection device, it may be a disposable, reusable or semi-reusable injection pen, an autoinjector or semi-reusable (or semi-disposable) autoinjector or a patch injector applicable onto the skin of the user for the duration of the injection. Reusable injection devices may include at least a part or a unit that can be reused for several injections. Semi-reusable devices include a unit or part that includes a disposable reservoir with the liquid drug which is discarded after an injection or if the reservoir is empty. The semi-reusable devices further include a reusable part typically including a dispensing mechanism.

[0029] Automatic devices or autoinjectors typically include automatic drive means such as an elastic element

(for example a preloaded spring) or an electric motor to drive a dispensing member to dispense the liquid drug from the reservoir.

[0030] The safety mechanism may include a counter element and wherein a first guiding surface of the trigger member or the counter element causes the trigger member or the counter element to move the lock element out of the locking position if the trigger member is moved along the longitudinal direction. The counter element may be part of or it may form the above mentioned intermediate member or alternatively the counter element may be a part of the housing of the delivery device.

[0031] If the first guiding surface contacts a contact element as a protrusion, cam or counter surface and if the trigger member is further moved along the longitudinal axis the first guiding surface is pressed onto the contact element the guiding surface slips along the counter element such that the longitudinal movement of the trigger member forces the counter element in a radial direction or in any direction inclined to the longitudinal direction

[0032] The trigger member may include the first guiding surface and the counter element includes a corresponding second guiding surface adapted to contact and engage the first guiding surface. An interaction of the first guiding surface with the second guiding surface causes the trigger member or the counter element to move the lock element out of the locking position.

[0033] The guiding surface may be a first sloped contact surface, it may be arranged on a protrusion, ledge or cam. The second guiding surface may be a second sloped contact surface.

[0034] The first and the second contact surface may be sloped in the same direction or have the same slope angle. The first and second sloped contact surface allow for a reliable actuating of the lock element. Due to the sloped surfaces the longitudinal movement of the trigger member can be reliably transferred to a radial movement or to a movement in an angled direction with respect to the longitudinal axis. In concrete terms, if the first sloped contact surface contacts the second sloped contact surface and if the trigger member is further moved along the longitudinal axis the first sloped surface is pressed onto the second sloped surface thereby causing the two surfaces to slip on each other such that the longitudinal movement of the trigger member forces the counter element in a radial direction or in any direction angled to the longitudinal direction.

[0035] The first and the second sloped contact surfaces may be positioned, for example, relative to the longitudinal axis in an angle of at least 20° or in an angle between 40° and 50° to the longitudinal axis.

[0036] The lock element may be a deflectable arm having an end connected to the needle cover, to the housing or to a holder and the arm has a free end part or end portion adapted to engage a stop surface or counter surface of the needle cover. Thus, if the needle cover is in the covering position the free end part or end portion of the arm engages or abuts the stop surface of the needle cover and thereby prevents a movement of the needle cover out of the covering position towards a retracted position. The lock element in form of a flexible arm allows an easy lock and unlock of the needle cover as the arm can be moved quickly out of the locking position.

[0037] Alternatively, the lock element may be a knob or ledge or the like.

[0038] As mentioned above the trigger member and the lock element are operatively coupled to each other. A movement of the trigger member in a first direction along the longitudinal axis may cause the lock element to move radially out of the locking position and a movement of the trigger sleeve in a second, counter direction along the longitudinal axis enables the lock element to move radially into the locking position. Thus, the lock element can be moved radially inwards or outwards, respectively depending on the longitudinal movement direction (distally or proximally) of the trigger member.

[0039] The term “move radially” includes movement that have at least a radial movement component. Hence, in case the lock element is a tongue or an arm these elements may swivel or pivot towards and away from the longitudinal axis and such a movement shall be encompassed by the term “move radially”.

[0040] In case the safety mechanism is used in a drug delivery device with an electric motor the motor can be driven forward and backwards to move the trigger member in either direction. Therefore, a simple and reliable locking or unlocking of the needle cover can be provided.

[0041] The drug delivery device may include a reservoir holder for holding the reservoir with the liquid drug. The needle cover may be movable relative to both the lock element and the reservoir holder if the needle cover is moved out of the covering position. In this embodiment the needle cover may be engaged and guided by the reservoir holder.

[0042] The safety mechanism may include a sleeve-shaped support member for holding the reservoir or a reservoir holder encompassing the reservoir. The support member may be arranged proximally or/and outside the trigger member and the support member may have a non-circular cross section. That provides a roll-away protection for the drug delivery device.

[0043] In a first embodiment the safety mechanism includes an intermediate member (or connecting element) which is may be a sleeve and which includes the counter element with a second guiding surface (or a second sloped contact surface). In this embodiment a movement of the trigger member along the longitudinal axis causes the first guiding surface (or first sloped contact surface) of the trigger member to interact with the second guiding surface (second sloped contact surface) of the intermediate member. This moves a portion of the intermediate member in a first radial direction or in an inclined direction thereby causing the lock element to move out of the locking position. The intermediate member may contact or abut the lock element and the lock element is moved by the intermediate member in a radial direction or inclined direction. Thus, the longitudinal movement of the trigger member is redirected to a radial movement of a portion of the intermediate member. This allows for a space-saving design as the trigger member is moved longitudinally and the intermediate member is moved radially. The dimensions of the safety mechanism in a radial direction are thus reduced compared to a mechanism with a radially moving trigger member.

[0044] In this embodiment the trigger member does not directly contact the lock element but the intermediate member is operatively positioned between the trigger member and the lock element. That means the lock element is controlled via intermediate member which allows to transfer a movement of the trigger member by means of a gear ratio or the intermediate member can be used to selectively

transfer or not transfer movements of the trigger member to the lock element. Therefore, a sophisticated kinematic control structure can be provided.

[0045] The trigger member with the guiding surface or first sloped contact surface may be exclusively moved along the longitudinal axis where the intermediate member with the second guiding surface or second sloped contact surface is exclusively moved in a radial direction or in a direction angled to the longitudinal axis. The intermediate member may abut, contact or engage the lock element to move it in or out of the locking position.

[0046] The intermediate member may be an arm, cam or protrusion. The intermediate member may be at least partially arranged coaxially inside the trigger member, it may be radially arranged between the lock element and the trigger member. This allows for a space-saving design.

[0047] The arm cam or protrusion may be connected to a sleeve-shaped member forming the intermediate member. In this embodiment the intermediate member has an asymmetric cross section in a plane perpendicular to the longitudinal axis. That means the intermediate member may have a movable portion or a movable element (in form of the mentioned arm, cam or protrusion) on one side or on several sides. The movable portion may be adapted to engage the trigger member and/or the lock element. The second guiding surface (second contact surface) may be arranged on or integrally formed in the movable portion of the intermediate member.

[0048] The trigger member may include further a third guiding surface and the intermediate member includes a fourth guiding surface. A movement of the trigger member in a second, counter direction along the longitudinal axis causes the third guiding surface to engage the fourth guiding surface and thereby moving a portion of the intermediate member in a second radial direction allowing the lock element to move from an unlocking or intermediate position into the locking position.

[0049] Therefore, by means of the third and fourth guiding surfaces the lock element can be easily and reliably moved into the locking position (thereby locking the needle cover) and by means of the first and second guiding surface the lock element can be moved out of the locking position (thereby releasing the needle cover).

[0050] As described above with respect to the first and second surface also the third and fourth surface may interact with each other such that the longitudinal movement of the trigger member is transferred to a lock element radial movement or a lock element movement in a direction angled to the longitudinal axis.

[0051] The third guiding surface may be implemented as third sloped contact surface and the fourth guiding surface may be a fourth sloped contact surface.

[0052] Instead of or additionally to the above the intermediate member (or a portion of the intermediate member) may be resilient or elastic such that the intermediate member can be deflected or biased and returns to an initial state due to the elastic properties.

[0053] The intermediate member (or a portion thereof) may be biased by the trigger member movement in the first longitudinal direction when the first guiding surface interacts with the second guiding surface and the intermediate member may return back or deflect back to the initial state

by the trigger member movement in the second longitudinal direction when the first and second guiding surfaces disengage from each other.

[0054] The biasing feature enables the intermediate member to switch back or move back to the initial state without additional interaction. This allows to ensure that intermediate member does not unintentionally move the lock element out of the locking position if, for example, the intermediate member is biased in a state in which the lock element is in the locking position.

[0055] The intermediate member may include a radially protruding actuation element in form of a cam, nose, lobe or protrusion. The control element is adapted to interact with and actuate the lock element. Depending on the form and the design of the actuation element a specific movement of the trigger member can be transferred to the lock element. Hence, the design of the actuation element defines how the lock element is actuated and how the lock element is moved by the intermediate member. The actuation element includes a contact portion which is adapted to contact the lock element to move the lock element into or out of the locking position.

[0056] Furthermore, the trigger member includes a control element wherein the first and third guiding surface (third contact surface) are arranged each on an opposite side of the control element in a longitudinal direction. The control element may be implemented as cam or protrusion.

[0057] The control element may protrude, for example, in a radial direction, e.g., in a plane perpendicular to the longitudinal axis. The control element may contact and engage the second and/or fourth guiding surface of the intermediate member if the trigger member is moved along the longitudinal direction. The control element allows a simple controlling and transfer of movement from trigger member to the intermediate member.

[0058] The second and fourth guiding surface may be arranged on an intermediate control element (cam or protrusion), which may be directed in a radial direction too. Therefore, the interaction between the trigger member and the intermediate member is implemented via control element and intermediate control element again allowing for a space-saving design.

[0059] The safety mechanism according to the disclosure includes in an embodiment a cap releasably attached at a distal end of the needle cover in an initial state or unused state of the drug delivery device. In this embodiment the needle cover or the cap includes a resilient and deflectable element (tongue, arm, protrusion or the like) holding the cap on the needle cover in a non-deflected state. A cap actuation element is operatively (direct or indirect) coupled to both the trigger member and the deflectable element. A movement of the trigger member in the longitudinal direction causes a movement of the cap actuation element which in turn causes a change of the deflectable element from the non-deflected state to a deflected state thereby releasing the cap from the needle cover. That means after the deflectable element is deflected the user can remove the cap from the needle cover to prepare the delivery device for an injection.

[0060] The deflectable element may be separate from the needle cover or from the cap or it may be integrally formed (monolithic) in the needle cover or in the cap. The deflectable feature allows to connect the cap easily by a releasable snap-fit connection.

[0061] In an embodiment the intermediate member includes the cap actuation element (e.g., the cap actuation element may be integrally formed in the intermediate member) and the trigger member includes further a first cap surface and the intermediate member includes a second cap surface which are both sloped surfaces, wherein the movement of the trigger member in the first direction along the longitudinal axis causes the first cap surface to engage the second cap surface thereby moving the intermediate member (or a portion thereof) in a first radial direction which causes the deflectable element to deflect.

[0062] The trigger member may include further a third cap surface and the intermediate member includes a fourth cap surface which are both sloped surfaces. A movement of the trigger member in a counter second direction along the longitudinal axis may cause the third cap surface to interact with the fourth cap surface thereby moving the intermediate member (or a portion thereof) in a second radial direction allowing the deflectable element to deflect back in its initial position.

[0063] The intermediate member includes in this embodiment not only an actuation element for moving the lock element but also a cap actuation element, for example in form of a protrusion, nose, lobe or cam for moving the deflectable element. That means the intermediate member provides two functions: actuating the lock element (for holding or releasing the needle cover) and actuating the deflectable element (for holding or releasing the cap). Hence, two function can be provided with one single member. For that purpose, the intermediate member may include two independently deformable portions, a first deformable portion for actuating the lock element and a second deformable portion for actuating the deformable element for the cap.

[0064] In a second embodiment of the safety mechanism according to the disclosure the housing includes the counter element. Thus, in contrast to the above mentioned first embodiment no intermediate member is present and the counter element is integrally formed in the housing or in a member fixedly connected to the housing.

[0065] The trigger member may include a body and a deflectable portion including the first guiding surface (sloped contact surface) and movement of the trigger member body in the first direction along the longitudinal axis causes the first guiding surface (sloped contact surface) to interact with the housing thereby moving the deflectable portion of the trigger member in a first radial direction or in a direction inclined with respect to the longitudinal axis thereby engaging and moving the lock element out of the locking position towards a non-locking position. Hence, the trigger member directly contacts and engages the counter element without any intermediate member.

[0066] Optionally, the housing includes a second guiding surface adapted to interact with the first guiding surface.

[0067] As described above with respect to the first embodiment when the lock element is moved out of the locking position the needle cover can be moved out of the covering position and thus exposing the needle. If the housing features the second guiding surface a simple design is provided as no further member or part is required. The first and second guiding surfaces may be a first and second sloped contact surface.

[0068] The deflectable portion of the trigger member may be, for example, implemented as arm, tongue or deflectable

protrusion. The deflectable portion may be essentially moved in a first radial direction or in a direction angled to the longitudinal axis by the longitudinal movement of the body of the trigger member.

[0069] In second embodiment the safety mechanism may include a cap releasably attached to a distal end of the needle cover in an initial state, shipping state or unused state of the delivery device. The cap or the needle cover includes a deflectable element holding the cap on the needle cover, it may be in a non-deflected state. A movement of the needle cover in the longitudinal direction out of the covering position towards a retracted position causes a change of the deflectable element from the non-deflected state to a deflected state thereby releasing the cap from the needle cover.

[0070] In an embodiment an automatic movement of the needle cover and cap relative to the housing releases the cap from the needle cover.

[0071] The cap can thus be hold on the needle cover by a releasable snap-fit connection. Furthermore, a simple longitudinal movement of the needle cover may release the connection and the cap can be removed and the delivery device is ready for an injection.

[0072] The disclosure relates further to an injection device for dispensing the liquid drug from the reservoir. The injection device includes a plunger rod and an automatic drive for driving the plunger rod to move a piston inside the reservoir to dispense the liquid drug form the reservoir through the injection needle or cannula. The injection device further includes the safety mechanism as described above.

[0073] The injection device may be, for example, a manual user-driven or automatic injection pen, an autoinjector including a spring element or motor for automatically driving the plunger rod or a semi-reusable injection device or semi-reusable autoinjector. Such reusable devices include a reusable drive unit and a disposable reservoir unit or syringe unit which is releasably attachable (for example by a thread or bayonet connection) to the drive unit.

[0074] The automatic drive may be adapted to rotate a drive member of the drive unit, where the drive member is operatively coupled (direct or indirect) to both the plunger rod and the trigger member, wherein rotation of the drive member by the automatic drive in a first rotational direction causes the plunger rod and the trigger member to be moved in the first direction along the longitudinal axis and rotation of the drive member in a second, counter rotational direction causes the plunger rod and the trigger member to be moved in the second direction along the longitudinal axis.

[0075] Hence, with one single automatic drive the plunger rod can be moved back and forth and the trigger member can be moved back and forth at the same time. That allows to lock and unlock the needle cover (via lock element) and, if any, lock and release a cap on the cover sleeve (via intermediate member, if any) and with the same automatic drive the plunger rod can be driven to dispense the drug form the reservoir. That means the automatic drive can be used to rotate the drive member to move the trigger member before an injection to lock or unlock the cover member and potentially the cap and subsequently the automatic drive can be used to dispense the drug during an injection or for preparation steps.

[0076] The drive member may be in treaded engagement with the plunger rod, such that the plunger rod is exclusively shifted along the longitudinal direction when the drive

member is rotated. The trigger member may be directly or indirectly coupled to the plunger rod such that a longitudinal movement of the plunger rod in the first direction causes to move the trigger member in the longitudinal direction in the first direction and movement of the plunger rod in the second longitudinal direction causes to move the trigger member in the second longitudinal direction.

[0077] The injection device further may include a reservoir unit (or syringe unit) and a drive unit. The reservoir unit may be releasably attachable to the drive unit. The reservoir unit includes a reservoir for the liquid drug (e.g., a cartridge, a syringe or a container), the needle cover and the lock element and the drive unit includes the housing, the automatic drive, the plunger rod, the trigger member and, if any, the intermediate member.

[0078] The reservoir unit may be disposable and the drive unit may be reusable. That means the injection device includes two units allowing to establish a semi-reusable concept which reduces the waste. In contrast to conventional injection devices the semi-reusable device does not have to be discarded completely but only the reservoir unit can be disposed after use or if the drug is fully dispensed out of the reservoir. The drive unit can be reused for several injections.

[0079] The reservoir unit is releasably attachable to the drive unit. That means the two units can be detached and reattached without destroying elements. Non-limiting examples of releasable connections are a snap-fit, a bayonet connection or a thread.

[0080] In an embodiment the reservoir unit includes a radially extending protrusion or radially extending rim portion and the drive unit includes a radially deflectable and flexible holding arm or clamping arm adapted to engage the protrusion or rim portion such that the syringe unit is held by the drive unit via snap-fit connection.

[0081] The drive unit may have additionally a lock to lock the snap-fit connection such that the syringe unit cannot be removed from the drive unit if the lock is activated. The lock may be implemented by the above described lock element or/and by a separate lock element preventing, for example, a radial movement of the holding arms and thereby locking the arms relative to the protrusion or rim portion.

[0082] The reservoir inside the reservoir unit may be a prefilled syringe with a non-releasably connected injection needle at the distal end thereof.

[0083] Upon attachment the reservoir unit may be held by the drive unit relative to the housing such that a movement of the reservoir unit in a dispensing direction is prevented.

[0084] The disclosure further refers to a method for a safety mechanism for an injection device for dispensing a liquid drug from a reservoir through a needle. The method includes the steps of

[0085] a. Providing an injection device with a housing defining a longitudinal axis;

[0086] b. Moving a trigger member inside the housing along the longitudinal direction

[0087] c. Moving a lock element, by the trigger member as the trigger member is operatively coupled to the lock element, out of a locking position in which the lock element engages a needle cover to prevent the needle cover from being moved out of a needle covering position,

[0088] d. Releasing, by the movement of the lock element, the needle cover from the covering position

and moving the needle cover out of the needle cover position towards a retracted position in which the needle is exposed.

[0089] A further aspect of the disclosure relates to an injection device for injection of fluid medicaments into a patient's tissue. Examples of possible injection devices are injection pens, autoinjectors, patch injection devices or the like. The injection device according to the disclosure includes electronics, with at least an electronic controller and a near field communication unit in functional connection to the controller. The injection device according to the disclosure further includes a cartridge assembly. The cartridge assembly has to be construed broadly. In one possible arrangement the cartridge assembly is mainly constituted by the cartridge as such, where the cartridge may be formed by a prefilled syringe, ampule or the commonly used carpule. In another arrangement the cartridge assembly may be constituted by a cartridge unit, e.g., including a housing, the cartridge as such, a needle protection sleeve, a needle assembly, and/or further constituents. It has to be understood that the cartridge assembly shall not be construed as a fully functional injection device of its own. So consequently, a driving arrangement shall not be part of the cartridge assembly even though particular elements of a driving arrangement may be located in the drive arrangement. The cartridge assembly includes in any case a near field communication readable item fixedly attached to the cartridge assembly. The near field communication readable item, in a simple design according to the disclosure, is a RFID or NFC (smart) tag (according to ISO/IEC 14443A/B, ISO/IEC 15693, ISO/IEC 18000-3, and/or according to any other suitable standard). Such tags may store information which may be read by a reading device. For example a tag according to the disclosure may store information regarding the medicament contained in the cartridge, the manufacturing date of the cartridge assembly and/or the medicament as well as any other information considered useful in the context of a therapy or in relation to the therapy. However, such tags may store a limited amount of data only, typically ranging from 48 Bytes to 1 Mbyte. For example, the NTAG213 by NXP (www.nxp.com) includes a user storage of 144 Bytes. According to the disclosure the near field communication unit of the injection device is capable of acquiring the data stored on the near field communication readable item. The cartridge assembly may be fixedly attached to the injection device. For example, the actual injection device may be manufactured at a first site, a prefilled syringe including the near field communication readable item (tag) as the cartridge assembly may be manufactured at a second site, and the end assembly, i.e. the mounting of the prefilled syringe, may take place at a third site. With the tag associated with the prefilled syringe it is not necessary to activate the injection device electronics during assembly, as all the necessary information may be obtained by the injection device at any point in time particularly also after assembly which simplifies the end assembly process significantly. Alternatively, the cartridge assembly may be releasably attachable to the injection device. In any case, if the cartridge assembly is attached to the injection device, the near field communication readable item and the near field communication unit are geometrically arranged at a distance to each other which allows for the stored information to be transferred from the item to the unit. The person skilled in the art is aware of appropriate geometrical arrangements for near field communication

arrangements. The near field communication unit according to the disclosure can be used in two modes. In a first mode, the near field communication unit is used to read information from the near field communication readable item. In the second mode, the near field communication unit is used to provide information to another device. The other device is a separate device, which includes a near field communication reading device. For example, the near field communication unit may read information from the near field communication readable item, provide the read information to the controller and subsequently change mode to provide at least a fraction of the read information and/or further data to the separate device. The information provided may be based on the read information or on information created by the controller.

[0090] In advantageous embodiments the near field communication device may use one single antenna for sending and receiving data.

[0091] In advantageous embodiments the injection unit includes only one wireless communication means, namely the near field communication unit, as described above.

[0092] In an embodiment the other device as described above may act as a display device for the injection device, rendering moot the need for an alphanumeric display, like an LCD or an OLED display, arranged on the injection device.

[0093] In a further developed example, the near field communication device of the injection device may be used not only to read information from the cartridge but also from the other device, which is e.g., a smart phone. For example the smart phone may receive recall information regarding the medicament used in the injection device and transfer information via near field communication containing e.g., a list of recalled lot number to the injection device. In case the lot number of medicament currently used in the injection device matches an entry of the list, the injection device may alert the user and/or block the injection device. The alert may be first transferred to the smart phone and be displayed on the display of the smart phone. Another possibility of a data transfer from the other device (smart phone) to the injection device might relate to the transfer of a kind of therapy plan from smart phone to injection device.

[0094] The controller may monitor operation of the injection device according to the disclosure or it may control at least part of the operation of the injection device according to the disclosure.

[0095] One aspect of the disclosure relates to an injection device as described where near field communication means near field communication (NFC) according to the standard ISO/IEC 18092 and its normative references or any of related standards by the IEEE organization or related.

[0096] One aspect of the disclosure relates to a system combining the injection device according to the disclosure and the separate device. The separate device may be a smart phone, a cellular phone, a personal digital assistant (PDA) like a palm pilot, a tablet computing device or any computing device with a near field communication reading device, like an NFC or a RFID reading device. The separate device runs AndroidOS, iOS by Apple, Windows operating system, Linux operating system, MacOS operating system, or any related operating system capable of operating the near field communication reading device.

[0097] In one aspect of the disclosure the information provided by the near field communication unit of the injection device to the separate device is data relating to state

information of the injection device or error information, both gathered by the controller. The error information may include error messages from the injection device as such, like operation failure, battery low state or the like. The error information may further or alternatively contain error information related to information acquired from the near field communication readable item, e.g., reached expiration date of the medicament, wrong medicament identification, or the like.

[0098] In one aspect of the disclosure the injection device according to the disclosure is a modular injection device. The modular injection device includes at least a reusable module and a disposable module which may be releasably attached to the reusable module. The reusable module contains at least the controller and the near field communication unit. Disposable and reusable modules are attached to each other by snap, bayonet, threaded connections or the like. The disposable module includes at least the cartridge assembly. After use or depletion of the cartridge, the disposable module is removed from the reusable module and disposed. A new disposable module may then be attached to the reusable module for further use.

[0099] In a more detailed example of the modular injection device according to the disclosure the injection device may be an autoinjector. The disposable unit is a cartridge unit. The reusable unit has a housing which houses at least a driving arrangement configured to force medicament out of a cartridge, electronics with a controller, and a near field communication unit. In, at, or on the housing a user interface is arranged which may be in functional communication with the controller. The cartridge unit includes a housing in which the cartridge with the fluid medicament is arranged. At a surface of the housing of the cartridge unit a near field communication readable item, for example a NFC tag, is arranged. The near field communication readable item stores information regarding the medicament, such as the type of medicament, the manufacturing date, information regarding shelf life and the like. Optionally, the tag may further include a temperature sensing element for logging of at least a maximum storage temperature or a humidity sensor.

[0100] The reusable unit's housing is configured to accommodate the cartridge unit at least partly within the housing with typically having the delivery end of the cartridge reaching out of a distal end of the reusable unit. The cartridge unit may releasably attached within the housing of the reusable unit for use. As the cartridge unit is accommodated in the reusable unit, the near field communication readable item of the cartridge unit gets located in the very proximity of the near field communication unit of the reusable unit. Driven by the controller, the near field communication may be used to read the information stored in the near field communication readable item. The read information may be used by the controller to influence operation of the injection device. Moreover, the information may be stored on the controller or a storage unit attached to the controller.

[0101] According to the disclosure the near field communication unit of the reusable unit of the injection device is not only configured to receive information but also to transmit information to other devices. In order to enable data transmission the controller is configured to set the near field communication unit into the so-called card emulation mode. Once in this mode the near field communication unit simulates a nearfield communication device from which infor-

mation may be read, similar to the near field communication readable item as described above. One possibility for another device which reads or receives information from the injection device may be a smart phone. The smart phone may be used to monitor a therapy and/or to give guidance to a person using the injection device.

[0102] In a further aspect the driving arrangement of the injection device includes an electric or electronic drive train including an element movable along an axis of the injection device which may be an autoinjector to force medicament out of the cartridge. For example, the movable element may be a piston rod applying force on a moveable piston in the cartridge, which in an example may be a prefilled syringe. The movable element may be driven by an electric motor, like a stepper motor, a DC motor, or piezoelectric motor. A gearing setup may interconnect the motor and the movable element.

[0103] In a further aspect, electrical and/or electronic components of an injection device according to the disclosure are energized through an energizing element like a battery, an accumulator or a super cap, wherein the energizing element of the injection device is disposed within a housing or a housing part of the injection device. The energizing element may be fixedly and non-removably disposed in the injection device. In an alternative the energizing element may be removably disposed in the injection device, either for recycling purposes or for exchange during use of the injection device.

[0104] In a further aspect, an injection device according to the disclosure includes a user interface. The user interface is operationally connected to the controller of the injection device. In a simple example the user interface consists of one or more LEDs (light emitting diodes), which may emit signals provided by the controller and meaningful to a person using the injection device. The signals may relate to a state of the injection device and/or to information received through near field communication from the cartridge or the cartridge unit.

[0105] In a more sophisticated example the user interface includes an LCD or OLED Display capable displaying more comprehensive information regarding a device state and/or information regarding the cartridge.

[0106] Various described aspects in this disclosure relate to near field communication. Near field communication with regards to the present disclosure shall be construed in its broadest possible sense as a wireless communication method using radio frequency electro-magnetic radiation to transmit information over a distance of less than one centimeter to several centimeters.

BRIEF DESCRIPTION OF THE DRAWINGS

[0107] The subject matter of the disclosure is explained in more detail in the following text with reference to exemplary embodiments which are illustrated in the attached drawings, in which:

[0108] FIG. 1 depicts an isometric view of a semi-reusable autoinjector, according to the disclosure;

[0109] FIG. 2 depicts an exploded view of the syringe unit of the autoinjector from FIG. 1;

[0110] FIG. 3 depicts a sectional view of the drive unit of the autoinjector;

[0111] FIG. 4 depicts the drive unit with an inserted syringe unit;

[0112] FIG. 5 depicts a sectional view of the autoinjector where the syringe unit is locked to the drive unit;

[0113] FIG. 6a depicts a syringe unit according to a second embodiment;

[0114] FIG. 6b depicts the syringe unit of FIG. 6a but without cap;

[0115] FIG. 7 depicts a sectional view of a drive unit of the second embodiment;

[0116] FIG. 8 depicts a perspective view of a gripper sleeve of the drive unit of the second embodiment;

[0117] FIG. 9a depicts the drive unit with an inserted syringe unit according to the second embodiment;

[0118] FIG. 9b depicts a sectional view of control cams for unlocking a cover sleeve and cap cams for unlocking a cap;

[0119] FIG. 10 depicts the cap cams in a locking state;

[0120] FIG. 11 depicts the control cams as well as the cap cams in a locking state;

[0121] FIG. 12 depicts the control cams and the cap cams during an injection;

[0122] FIG. 13 depicts the autoinjector after release of the syringe unit and

[0123] FIG. 14 depicts a third embodiment of the autoinjector;

[0124] FIG. 15 depicts a syringe unit configured to be inserted into a reusable drive unit of the injection device which may communicate with a smart phone;

[0125] FIG. 16 depicts the injection device with its outer distal housing removed revealing the near field communication unit; and

[0126] FIG. 17 depicts a longitudinal cross section of the injection device showing the syringe unit with the prefilled syringe. The further elements present in the injection device except the housing element are omitted.

[0127] The reference symbols used in the drawings, and their primary meanings, are listed in summary form in the list of designations. In principle, identical parts are provided with the same reference symbols in the figures.

DETAILED DESCRIPTION

[0128] The term “injection device” or “injector” refers to a device that is removed from the injection site after each medication event or drug delivery process. This is in contrast to the term “infusion device” which refers to a device with a cannula or needle that remains in the skin of the patient for a prolonged period of time, for example, several hours.

[0129] In the present context, the terms “substance”, “drug”, “medicament” and “medication” are to be understood to include any flowable medical formulation suitable for controlled administration through a means such as, for example, a cannula or a hollow needle, and includes a liquid, a solution, a gel or a fine suspension containing one or more medical active ingredients. A medicament can be a composition including a single active ingredient or a pre-mixed or co-formulated composition with more than one active ingredient present in a single container. Medication includes drugs such as peptides (e.g., insulin, insulin-containing drugs, GLP-1 containing drugs or derived or analogous preparations), proteins and hormones, active ingredients derived from, or harvested by, biological sources, active ingredients based on hormones or genes, nutritional formulations, enzymes and other substances in both solid (suspended) or liquid form but also polysaccharides, vaccines,

DNA, RNA, oligonucleotides, antibodies or parts of antibodies but also appropriate basic, auxiliary and carrier substances.

[0130] The term “distal” refers to the direction or the end of the drug delivery device carrying an injection needle or an injection cannula, whereas the term “proximal” is meant to refer to the opposite direction or end pointing away from the needle or cannula. In the following description the radial direction is meant to refer to a direction perpendicular to the longitudinal direction or longitudinal axis. The latter is defined by the housing and refers to the direction in which the housing (and the delivery device) has the longest dimension. In the present description the term “distal” refers to the side where the injection needle is located. This is on the left hand side in the FIGS. 1 to 13. The term “proximal” refers to the opposite side or rear end of the device and is on the right hand side in FIGS. 1 to 13.

[0131] First embodiment: FIG. 1 shows a perspective view of a drug delivery device of a first embodiment of the disclosure in form of a semi-reusable autoinjector 1. The autoinjector 1 includes a disposable syringe unit 2 and a reusable drive unit 3. As shown in FIG. 1 the drive unit 3 has an elongated housing defining a longitudinal axis and includes a button 4 at its proximal end. In the following, the structural features of the syringe unit and the drive unit will be described in detail. Subsequently, the function of the semi-reusable autoinjector 1 will be explained.

[0132] FIG. 2 depicts an exploded view of the syringe unit 2. As shown in FIG. 2 the syringe unit 2 includes a syringe holder 30, a cover sleeve 20 coaxially arranged around the syringe holder 30 and movable relative to the syringe holder along the longitudinal axis and a prefilled syringe 15 including a liquid drug and a rigid needle shield 17 on the injection needle. The cover sleeve includes a label 25 wrapped around the cover sleeve 20. Further, a cap 10 is arranged on a distal end to cover the injection needle. The syringe holder 30, the cover sleeve 20 and the cap 10 are individually injection-molded plastic parts. The cap 10 may include additional metallic parts.

[0133] The prefilled syringe 15 with a barrel made of glass is rotationally and axially fixed inside the sleeve-shaped syringe holder 30. For that purpose, the syringe holder 30 includes clamping elements pressing on an outer surface of the syringe barrel and a distal shoulder of the syringe abuts a bearing surface 34 (see FIG. 4) inside the syringe holder 30. As shown in FIG. 2 the syringe holder 30 further features longitudinal guiding rails 31 on its outside for the cover sleeve 20 such that the cover sleeve 20 can be shifted exclusively in the longitudinal direction relative to the syringe holder 30. Furthermore, the syringe holder 30 includes two radially deflectable lock elements in form of locking arms 32 arranged on opposite sides of the syringe holder 30. In an initial state the locking arms 32 are in a non-deflected state and in a locking position and prevent a movement of the cover sleeve 20 relative to the syringe holder 30 in a proximal direction. Hence, the cover sleeve 20 shields or covers the injection needle of the syringe 15 in a covering position. For that purpose, a free end of the locking arms 32 abut stop surfaces 24 inside the cover sleeve 20. As will be described below once the locking arms 32 are deflected radially inwards the cover sleeve 20 can be pushed proximally. The syringe holder 30 further includes two oppositely arranged and radially protruding ledges 33 at a

proximal end of the syringe holder 30 for engagement with clamping arms 61 (FIG. 4) inside the drive unit 3.

[0134] The cover sleeve 20 has a non-circular cross section in a plane perpendicular to the longitudinal axis. That means two sides of the cover sleeve are flattened. The cover sleeve includes a first lateral opening 21 for access to the locking arms 32 and a proximal second opening 22 for access to the ledges 33 from outside. The label 25 wrapped around the cover sleeve 20 includes a machine-readable tag in form of a RFID or QR code 26 identifying the syringe unit 2.

[0135] At a distal end of the cover sleeve 20 the cap 10 is releasably attached by a snap-fit connection. The cap 10 includes on two sides two deflectable elements or tongues 11 and on each side a protrusion engaging in a corresponding indentation 23 in the cover sleeve 20 for the snap-fit connection. Alternatively, the cap 10 may have an ellipse shape in a plane perpendicular to the longitudinal axis wherein inside the ellipse form the protrusions engage the indentation in the cover sleeve. The ellipse can be deformed to a circular shape to release the protrusions from the indentation 23. The release of the snap-fit will be described below with respect to the de-capping of the syringe unit 2. The cap 10 includes in its inside a sleeve-shaped holder 12 (FIG. 2) connectable to the rigid needle shield (RNS) 17 of the syringe 15 such that the RNS 17 is removed with the cap 10 if the cap is released from the needle cover.

[0136] The syringe 15 inside the syringe holder 30 has a barrel made of glass and inside the barrel a movable piston 16 (FIG. 4). An injection needle 18 (FIG. 4) is non-releasably connected to a distal end of the syringe barrel. In an unused state the injection needle 18 is covered by the RNS 17. The latter in turn is axially and radially engaged or engageable by the holder 12 in an assembled or ready-to-use state of the syringe unit.

[0137] FIG. 3 depicts a sectional view of a first embodiment of the drive unit 3. The cut of the sectional view runs along the longitudinal axis of the device. The drive unit 3 includes a housing 80 including an outer housing cover and inside the cover a sleeve mechanism 83 including a radial wall 84 that separates the drive 3 unit in a distal portion and a proximal portion. The mechanics sleeve 83 is immovably connected to the housing 80. Inside the housing 80 and distally the separating wall 84 is a distal housing insert 81 whereas proximal the wall 84 a proximal housing insert 85 is located. The distal and proximal housing inserts 81, 85 are immovably connected to the housing 80. Furthermore, the drive unit 3 includes an opening on its distal side adapted to accommodate a proximal portion of the syringe unit 2.

[0138] Inside the distal housing portion and guided by the distal housing insert 81 a gripper sleeve 60 is arranged and movable relative to the housing 80 along the longitudinal axis. Coaxially to the gripper sleeve 60 a trigger sleeve 50 is arranged. Again inside the trigger sleeve 50 a cover sleeve connector 45 is located and biased in distal direction by a cover sleeve spring 46. The spring abuts on its distal end the cover sleeve connector 45 and on its proximal end the radial wall 84 of the sleeve mechanism 83. A sleeve-shaped syringe connector 47 is coaxially arranged inside the cover sleeve connector 45 and biased distally by a syringe connector spring 48 coaxially inside the cover sleeve spring 46 and proximally supported by the wall 84. The spring 46 ensures that the syringe connector 45 pushes the syringe 15 of an inserted syringe unit 2 in distal direction.

[0139] The proximal housing portion accommodates the proximal housing insert **85** which supports and guides a sleeve-shaped trigger sleeve connector **51** movable along the longitudinal axis, a drive assembly with an electric motor **91**, a pinion **92** and a gear **93** connecting the motor **91** with a threaded rod **41**, a battery **94**, the threaded rod **41** and a plunger rod **40** threadedly connected to the threaded rod **41**. The proximal housing portion further accommodates an electronic module with a controller configured to control the electric motor **91** and providing information to the user via a display or LEDs or a communication module. The trigger sleeve connector **51** is fixedly and immovably connected to the trigger sleeve **50** and distally pressed onto a proximal end of the plunger rod **40** by a trigger sleeve spring **53**. The plunger rod **40** non-rotatably guided by the sleeve mechanism **83** and includes on its distal end a flange **49** adapted to engage the piston **16** (see FIG. 4) during dispensing.

[0140] Inside the proximal housing portion is further an axially arranged detection pin axially movable by the cover sleeve connector **45**. The detection pin is connected to and interacts with a TMR position sensor. The presence of a syringe unit **2** inside the drive unit housing **80** is thus detectable via detection pin and the position sensor which provides a corresponding signal to the controller of the electronic module.

[0141] The gripper sleeve **60** includes two axially extending and deflectable clamping arms **61** having a clamp **62** at their free distal end adapted to accommodate the syringe holder ledges **33**. The clamping arms **61** can be deflected radially outwards allowing the clamps **62** to snap onto the ledges **33** of the syringe holder **30** when the syringe unit is inserted into the opening in the drive unit **3**. On its outside the gripper sleeve **60** includes splines running in corresponding longitudinal grooves in the distal housing insert **81** guiding the gripper sleeve **60** during an axial movement relative to the housing **80**. Furthermore, each clamping arm **61** provides a support for an axially aligned gripper sleeve spring **64** which biases the clamping arms **61** in a holding position.

[0142] The trigger sleeve **50** is axially guided by the gripper sleeve **60** as well as by the distal housing insert **81** and shiftable relative thereto. For this purpose, the trigger sleeve **50** includes longitudinal guiding rails engaging longitudinal grooves in both the distal housing insert **81** and the gripper sleeve **60**. As it can be seen in FIG. 3 the trigger sleeve **50** includes oppositely arranged trigger arms **54** which include on their proximal free end **56** a guiding surface or sloped contact surface **55**. The trigger arms **54** are deflectable radially inwards upon contact of the contact surface **55** with a counter guiding surface or counter sloped contact surfaces **82** of the distal housing insert **81**. As mentioned above the trigger sleeve **50** is further immovably connected to the trigger sleeve connector **51** in the proximal housing portion. As the trigger sleeve connector **51** is biased by the trigger sleeve spring **53** the trigger sleeve **50** is biased too in distal direction. The trigger sleeve connector **51** in turn is connected to the plunger rod via ledges **52** which engage a proximal shoulder of the plunger rod **40** such that an axial movement of the plunger rod **40** in proximal direction moves also the trigger sleeve **50** proximally.

[0143] In the state shown in FIG. 3 the drive unit **3** is ready to be loaded with a syringe unit **2**. The controller displays the

current state to the user via display by showing a message like “Syringe Unit Insertion” or via green or red LEDs.

[0144] In order to prepare the autoinjector **1** for an injection the user attaches a syringe unit **2** to the drive unit **3**. The electronics in the drive unit **3** are switched from an inactive state or sleep mode to an active mode. Alternatively, the user can press the button **4** to activate the electronics.

[0145] FIG. 4 depicts a sectional view of the drive unit **3** with an inserted syringe unit **2**. The syringe unit **2** has been inserted with its proximal end portion into the distal opening of the drive unit housing **80**. In the state shown in FIG. 4 the proximal ledges **33** of the syringe holder **30** are snapped into recesses of the clamps **62** of the gripper sleeve clamping arms **61**. The syringe unit **2** is thus held in the drive unit **3** by the clamping arms **61**. However, in the shown state in FIG. 4 the syringe unit **2** can still be pulled out of the drive unit housing by overcoming a holding force. Hence, if the user pulls the syringe unit **2** in a distal direction the syringe holder **30** with the syringe **15** are firstly moved in distal direction relative to the cover sleeve **20** until stop and subsequently the cover sleeve **20** is moved distally too. This brings inclined release surfaces **27** of the cover sleeve **20** into contact with inclined counter surfaces **65** of the clamping arms **61**. If the cover sleeve **20** is further pulled in distal direction the clamping arms **61** are deflected radially outwards via release surfaces **27** and counter surfaces **65**. The ledges **33** of the syringe holder **30** are therefore released from the clamping arms **61** and the syringe unit **2** can be removed from the drive unit **3**, out of the drive unit **3** the clamping arms **61** may deflect radially outwards and release the ledges **33**.

[0146] That means in the state shown in FIG. 4 the syringe unit **2** is held inside the drive unit **3** but not yet locked inside the drive unit **3**.

[0147] FIG. 5 depicts the autoinjector **1** in a locked state without cap. When the syringe unit **2** is inserted as described above the syringe unit **2** moves the detection pin and thus the position sensor detects the movement and automatically sends a signal to the controller in the electronic module which detects the presence of the syringe unit **2** inside the drive unit housing. A code reader in form of an NFC reader or QR code reader inside the drive unit **3** reads the RFID or QR code **26** of the syringe unit **2**. Subsequently, the controller of the drive unit sends the read medication information from the read code **26** to an external device to verify the drug or medication with predefined information. Alternatively, the controller compares the read medication information from the code **26** with predefined information stored in a memory inside the drive unit **2**.

[0148] In case of successful medication verification the controller receives an enable signal (either from the external device or from the internal comparison). The controller then controls the electric motor **91** to rotate the threaded rod **41** to move the trigger sleeve **50** a short distance of several millimeters in proximal direction. Such a trigger sleeve **50** movement is achieved by a kinematic chain as described as follows.

[0149] The motor **91** rotates the pinion **92** and the gear **93** transfers the rotation to the threaded rod **41** which in turn moves the non-rotating plunger rod **40** in proximal direction. As the trigger sleeve connector **51** is connected via its ledges **52** to a proximal shoulder of the plunger rod **40** the latter causes the trigger sleeve connector **51** to move and thus the trigger sleeve **50** is shifted in proximal direction too.

[0150] The proximal movement of the trigger sleeve 50 brings the sloped contact surfaces 55 of the trigger arms 54 into contact with the counter sloped contact surfaces 82 of the distal housing insert 81. As the two sloped surfaces 55, 82 slide along each other the trigger arms 54 are forced to deflect radially inwards when the trigger sleeve 50 is further moved in proximal direction. The free ends 56 of the deflected trigger arms 54 in turn come into contact with the locking arms 32 of the syringe holder 30. Subsequently, the trigger arms 54 deflect the locking arms 32 radially inwards and thus move the locking arms 32 out of a locking position. As a consequence, the locking arms 32 no longer engage the stop surface of the cover sleeve 20 and thus allow for the cover sleeve 20 to move out of a covering position and move inside the housing 80 in proximal direction relative to the syringe holder 30 and relative to the drive unit housing 80.

[0151] At the same time the radially inwards deflected trigger sleeve arms 54 abut a distal stop surface of the syringe holder 30 and thereby lock the syringe holder 30 relative to the trigger sleeve 50 such that a movement of the trigger sleeve 50 is transferred to the syringe unit 2. Thus, in this state (FIG. 5) the syringe unit 2 can no longer be pulled out of the drive unit 3.

[0152] The controller displays the locked state to the user via display. As in this state the cap is still mounted onto the distal end of the cover sleeve 20 the next step is the de-capping which may be started by pressing the button 4. The controller then drives the motor 91 to move additionally the trigger sleeve 50 a short distance of 1 to 5 mm in proximal direction and immediately the same distance back in distal direction. As the trigger arms 54 fix the syringe unit 2 relative to the trigger sleeve 50 the syringe unit 2 (and the gripper sleeve 60) are moved back and forth too. The proximal movement of the syringe unit 2 with the cap 10 relative to the drive unit housing 80 compresses the trigger sleeve spring 53. In FIG. 5 it can be seen the trigger sleeve connector 51 was moved a distance in proximal direction (see distance between ledges 52 and proximal end of sleeve mechanism 83). The proximal movement of the syringe unit 2 relative to the housing 80 further causes the deflectable cap tongues 11 (FIG. 2) of the cap 10 to abut against a triangle-shaped protrusion 86 (FIG. 3) on a distal end of the drive unit housing 80. As a consequence, if the syringe unit 2 is further moved into the drive unit housing 80 the protrusion 86 deflects the cap tongues 11 and thereby releases the snap-fit connection of the cap 10 with the cover sleeve 20. Consequently, the cap 10 is wiped off. This leads to the state shown in FIG. 5. The controller moves then motor 91 in counter direction to move the plunger rod 40 back the short distance of 1 to 5 mm but without unlocking the syringe unit 2. The syringe unit 2, the trigger sleeve 50 and the gripper sleeve 60 are moved back due to the spring force of the trigger sleeve spring 53.

[0153] The autoinjector 1 is now de-capped and ready for an injection. The controller displays a corresponding notification on the display.

[0154] As a next step the user places the distal end of the cover sleeve 20 onto an injection site and pushes the autoinjector 1 towards the injection site. This causes the cover sleeve 20 to move proximally from the distal covering position into the drive unit housing 80 (push on skin) and compresses the cover sleeve spring 46 which biases the cover sleeve 20 in distal direction via cover sleeve connector 45. Upon push on skin the cover sleeve connector 45 is

moved proximally and thus the detection pin is further moved proximally. This further movement is detected by a further TMR position sensor which triggers the controller in the electronic module to start the injection. That means the controller drives the electric motor 91 in dispensing direction and thus rotates the threaded rod 41 and thereby moves the plunger rod 40 in distal direction. As the trigger sleeve spring 53 biases the trigger sleeve connector 45 on the proximal shoulder of the plunger rod 40 (alternatively the trigger sleeve connector 45 is immovably connected to the plunger rod 40) the trigger sleeve 50 is also moved in distal dispensing direction. The distal flange 49 of the plunger rod 40 moves the piston 16 inside the syringe 15 in distal direction and thus dispenses liquid drug through the injection needle out of the syringe 15.

[0155] The distal movement of the plunger rod 40 moves the trigger sleeve via trigger sleeve connector 51 in distal direction to its initial position. That means the trigger arms 54 can move radially outwards back into their non-deflected position. However, as the cover sleeve 20 is in a retracted proximal position the locking arms 32 are prevented from being moved back into the locking position.

[0156] The controller displays a holding time indicating a time period during that the user has to hold the autoinjector 1 onto the injection side after the injection to ensure that the drug is completely administered and absorbed. The controller may display a down counter informing the user about the remaining holding time. Once the holding time is elapsed a notification is shown to the user that the autoinjector can be removed from the injection site. Alternatively, the end of the holding time may be indicated only by a LED.

[0157] When the user removes the autoinjector 1 from the injection site compressed cover sleeve spring 46 can release and thus it can move the cover sleeve 20 back distally into the needle covering position.

[0158] During dispensing the trigger sleeve 50 was moved in its initial distal and unlocking position in which the trigger arms 54 are not deflected anymore and thus do not contact the locking arms 32 of the syringe holder 30. That means the locking arms 32 are no longer completely pressed radially inwards and upon movement of the cover sleeve 20 into its covering position the locking arms 32 are able to switch back into their locking position in which the free end of the arms 32 abuts the contact surface inside the cover sleeve 20 thereby locking the cover sleeve 20 in its covering position as in the state shown in FIG. 4.

[0159] Furthermore, as the trigger arms 54 are not deflected anymore and thus do not contact the distal stop surface of the syringe holder 30 the entire syringe unit 2 can be pulled out of the drive unit housing 80 by the user. The user can now discard the syringe unit 2 and the drive unit 3 is ready to be loaded with a new syringe unit.

Second embodiment

[0160] In the following a second embodiment of the present disclosure is described with respect to FIGS. 6a to 13. The drug delivery device of the second embodiment is a semi-reusable autoinjector 100 including a drive unit 103 and a syringe unit 102. The differences to the first embodiment autoinjector will be described in detail.

[0161] The syringe unit 102 of the second embodiment is shown in FIGS. 6a and 6b is similar to the syringe unit of the first embodiment. FIG. 6a depicts a perspective view of

the syringe unit 102 with the cap 110 and FIG. 6b depicts the syringe unit 102 without cap.

[0162] In contrast to the first embodiment the syringe holder 130 of the syringe unit 102 of the second embodiment includes two cap holding tongues 135 extending axially and deflectable radially inwards. The two oppositely arranged tongues 135 have each a free end including a radially outwards extending protrusion 136 adapted to interact with a cap actuation nose 179 as described below. Each tongues 135 further has two ledges 137 next to protrusion 136 on both sides. The ledges 137 engage corresponding recesses in the cap 110 to hold the cap 110 on the cover sleeve 120 by a releasable snap-fit connection.

[0163] FIG. 7 depicts a sectional view of a drive unit 103 of the second embodiment. In contrast to the first embodiment described above the drive unit 103 includes an intermediate sleeve 160 radially positioned between a distal housing insert 181 and a trigger sleeve 150.

[0164] FIG. 7 depicts a perspective view of the intermediate sleeve 160. As shown in FIG. 7 the intermediate sleeve 160 includes two axially extending levers 170 that are connected to a sleeve-shaped element similar to the above described gripper sleeve of the first embodiment. The intermediate sleeve 160 further includes two clamping arms 161 (see FIG. 7) for holding the syringe unit 102 as described with respect to the first embodiment. Each clamping arm 161 is connected to a spring 162. The spring 162 is supported on a sleeve mechanism 183 and biased the clamping arms 161 in a holding position. The intermediate sleeve 160 is axially fixed relative to the housing 180 and to the distal housing insert 181.

[0165] Furthermore, as shown in FIGS. 7 and 8 the intermediate sleeve 160 includes on its levers 170 two radially inwards protruding actuation noses 178 adapted to abut locking arms 132 of an inserted syringe unit 102. Additionally, on a distal end each lever 170 includes a radially inwards protruding cap actuation nose 179 which are adapted to interact with the deflectable cap holding tongues 135 of the syringe unit 102. The trigger sleeve 150 includes lateral openings allowing the actuation nose 178 and the cap actuation nose 179 to extend through to reach the locking arms 132 and the tongues 135, respectively. The levers 170 include radially outside the actuation nose 178 counter cams 171 extending in a plane perpendicular to the longitudinal axis and including contact surfaces on a distal and on a proximal side as will be described in detail below. Correspondingly, the levers 170 include radially outside the cap actuation noses 179 counter cap cams 175 with contact surfaces. The contact surfaces of the counter cam 171 and of the counter cap cams 175 are adapted to interact with contact surfaces of control cams 151 and cap cams 155 of the trigger sleeve 150.

[0166] FIG. 9a shows the autoinjector 100 in an unlocked state in a sectional view wherein the cut runs along the longitudinal axis. The syringe unit 102 is inserted into the drive unit 103 but not yet locked to the drive unit 103. FIG. 9b shows the same state but the cut for the sectional view is offset parallel to the cut of FIG. 9a to show the control cams 151, cap cams 155 and counter cams 171 and counter cap cams 175.

[0167] As shown in FIG. 9b the trigger sleeve 150 control cams 151 are arranged on an outer portion oppositely to each other and extending in a plane perpendicular to the longitudinal axis. The control cams 151 are adapted to interact

with the above mentioned counter cams 171 of the intermediate sleeve 160. Additionally, the cap cams 155 of trigger sleeve 150 are arranged on a distal end portion oppositely to each other and are adapted to interact with the corresponding counter cap cams 175 of the intermediate sleeve 160.

[0168] Further as shown in FIG. 9b the control cams 151 include on a distal side a first sloped contact surface 152 adapted to engage a second sloped contact surface 172 on a proximal side of the counter cam 171 of the intermediate sleeve 160. Furthermore, the control cam 151 includes on a proximal side a third sloped contact surface 153 adapted to engage a fourth sloped contact surface 173 on a distal side of the counter cam 171. As shown in FIG. 9b all sloped contact surfaces 152, 153, 172, 173 have the same angle to the longitudinal axis.

[0169] Similarly, the cap cams 155 include a distal surface 156 and a proximal surface 157 and the counter cap cams 175 of the intermediate sleeve 160 include a counter distal surface 176 and a counter proximal surface 177. As it can be seen in FIG. 9b the distal and proximal surfaces of the cams 155, 175 are oppositely angled such that the cap cams 155 and counter cap cams 175 are shaped as a truncated pyramid in a plane parallel to the longitudinal axis.

[0170] The function of the second embodiment autoinjector 100 will be described in the following.

[0171] In a first step, the user inserts a proximal portion of the syringe unit 102 into the housing of the drive unit 103. Upon contact with the syringe unit ledges 133 the clamping arms 161 deflect radially and snap onto the proximal ledges 133 of the syringe holder 130 as described in the first embodiment. The syringe connector 147 biased by the syringe connector spring 148 abuts a proximal rim portion or flange of the syringe 15 and thus ensures that the syringe is pressed against a bearing surface inside the syringe holder 130.

[0172] The position sensor detects that the syringe unit 102 has been inserted and provides a corresponding signal to the controller. Subsequently, after a successful medication verification as described above with respect to the first embodiment the controller drives the electric motor 191 to automatically retract the trigger sleeve 150 via trigger sleeve connector 158 in a proximal direction.

[0173] That means the trigger sleeve 150 is axially moved relative to the intermediate sleeve 160 which does not move. Therefore, the control cams 151 and the cap cams 155 of the trigger sleeve 150 move from an initial distal position as shown in FIG. 9b to a proximal position as shown in FIG. 10. During the proximal movement of the trigger sleeve 150 the control cams 151 thus pass radially inwards the counter cams 171 of the intermediate member 160. The cap cams 155, however, come into contact with their proximal sloped surface 157 with the distal sloped surface 176 of the counter cap cam 175 during trigger sleeve 150 movement relative to the intermediate sleeve 160.

[0174] Due to the movement of the trigger sleeve 150 the cap cam proximal sloped surface 157 slide along the counter cap cam distal surface 176 and thereby forcing the levers 170 of intermediate member 160 to deflect radially inwards which moves the cap actuation noses 179 radially inwards too. As the trigger sleeve 150 is further shifted proximally the cap cams 155 of the trigger sleeve passes above the counter cap cam 175 and thus keeps the levers 170 in a radially inwards deflected position as shown in FIGS. 10 and 11.

[0175] In case the medication verification is not successful (e.g., an incorrect syringe unit was inserted) the controller drives the motor to move the trigger sleeve 150 in distal direction to release the syringe unit 102 from the drive unit 103 as described below with respect to the last step after an injection.

[0176] As the cap actuation noses 179 are now moved radially inwards the noses 179 press onto the protrusion 136 of the cap holding tongues 135 and thus deflect the holding tongues 135 radially inwards. This releases the ledges 137 from the cap 110 and thus the snap-fit connection between the tongues 135 and the cap 110 is released and the user can remove the cap 110 from the cover sleeve 120.

[0177] Subsequently, a cover sleeve lock is unlocked and the syringe unit 102 is locked to the drive unit 103. That is, the controller drives the motor 191 in dispensing direction causing the plunger rod 140 to move in distal direction. As the trigger sleeve connector 158 is pressed onto a proximal end of the plunger rod 140 by the trigger sleeve spring 159 (or alternatively the trigger sleeve connector is fixedly connected to the plunger rod) the trigger sleeve 150 is moved via trigger sleeve connector 158 in distal direction too.

[0178] This brings the first contact surfaces 152 of the control cams 151 into contact with the second contact surfaces 172 of the counter cams 171 of the intermediate member 160. As the first and second contact surfaces 152, 172 are sloped the two surfaces slide along each other and as the trigger sleeve 150 is axially guided and cannot move in radial direction the control cams 151 press the counter cams 171 radially inwards such that control cams 151 pass radially outwards the counter cams 171 (FIG. 11). This bends or deflects a middle portion of the levers 170 of the intermediate sleeve 160 radially inwards and thereby moves the actuation noses 178 radially inwards as shown in FIG. 11. As a consequence, the actuation noses 178 press on the deflectable locking arms 132 of the syringe holder 130 thereby move the locking arms 132 radially inwards, see FIG. 11. This in turn moves the free end of the locking arms 132 out of a locking position and the arms 132 no longer contact the stop surface of the cover sleeve 120. The cover sleeve 120 is thus unlocked and is able to move proximally relative to the syringe holder 130 and relative to the housing 180.

[0179] At the same time and in a manner analogous as in the first embodiment the radially inwards deflected actuation noses 178 abut a distal stop surface of the syringe holder 130 and thereby lock the syringe holder 130 relative to the trigger sleeve 150 such that a movement of the trigger sleeve 150 is transferred to the syringe unit 2. Thus, in this state syringe unit 102 is locked inside the drive unit 3.

[0180] During the above described distal movement of the trigger sleeve 150, the cap cams 155 are moved relative to the counter cap cams 175 of the intermediate sleeve 150. However, the cap cams 155 still press the counter cap cams 175 radially inwards and thus keep the cap actuation noses 179 in the radially inwards position as depicted in FIG. 11.

[0181] The controller displays a message on the display or activates LEDs to inform the user that the autoinjector is ready for and injection.

[0182] FIG. 11 shows the above described state of the autoinjector 100 where the cap 110 is released and the cover sleeve 120 is unlocked and the syringe unit is locked by the radially inwards moved actuation noses 178.

[0183] As described with respect to the first embodiment the injection is started by a push on skin release. FIG. 12 depicts the state where the cover sleeve 120 is moved proximally out of the covering position if the user presses the autoinjector 100 onto the skin. The cover sleeve 120 moves the cover sleeve connector 145 proximally and thereby compressing the cover sleeve spring 146. In a proximal end position a TMR position sensor detects via detection pin the proximal position of the retracted cover sleeve connector 145 and provides a start signal to the controller. The latter starts the injection by driving the motor 191 in dispensing direction.

[0184] The motor 191 rotates the threaded rod 141 and moves the non-rotating plunger rod 140 in distal direction and thus moves the trigger sleeve 150 via trigger sleeve connector 158 in distal direction. That means the trigger sleeve 150 with its control cams 151 and cap cams 155 is moved relative to the intermediate sleeve 160. The control cams 151 and cap cams 155 are moved from the position shown in FIG. 11 to the position shown in FIG. 12 wherein the control cams 151 and cap cams 155 pass and release the counter cams 171, 175 of the intermediate sleeve 160 and thus allow the intermediate sleeve levers 170 to deflect back into their initial non-deflected state. Thereby the actuation noses 178 and cap actuation noses 179 are move radially back outwards into their initial state. That would in turn return the syringe holder locking arms 132 towards their locking position. However, as the cover sleeve 120 is in a proximal retracted position the locking arms 132 cannot move completely radially outwards.

[0185] The motor 191 further drives the plunger rod 140 in distal dispensing direction and the plunger rod 140 moves the piston 16 inside the syringe 15 distally to dispense the liquid drug until the drug is completely dispensed. Subsequently, the controller displays a holding time counter as described with respect to the first embodiment.

[0186] When the user removes the autoinjector 100 from the injection site the cover sleeve 120 can move back into the covering position by a force of the compressed the cover sleeve spring 146. As the levers 170 of the intermediate sleeve 160 and thus the actuation nose 178 and the cap actuation nose 179 are in the retracted state the locking arms 132 of the syringe holder 130 can move back radially outwards once the cover sleeve 120 is moved back distally in its covering position. The cover sleeve 120 is therefore locked again by the locking arms 132 and thus is prevented from a movement in proximal direction.

[0187] The distal movement of the cover sleeve 120 moves the cover sleeve connector 145 too and the position sensor detects the movement via detection pin. Upon confirmation of the user via button the controller releases the lock of the syringe unit 102. For this purpose, the controller rotates the motor 191 in reverse direction to move the trigger sleeve 150 in a proximal end position. The proximal movement of the trigger sleeve 150 brings a proximal (third) contact surfaces 153 of the control cams 151 into contact with distal (fourth) contact surfaces 173 of the counter cams 171. Due to the sloped angle the counter cams 171 are moved radially outwards and thus move the levers 170 into their initial position (if the levers 170 are not already fully deflected back in the initial position). The control cams 151 and the cap cams 155 then pass the counter cams 171 and counter cap cams 175 respectively to a proximal position such that the cams of the trigger sleeve 150 do not contact

the counter cams of the intermediate sleeve 160 anymore and thus levers 170 are not deflected any more.

[0188] If the trigger sleeve 150 is moved towards its proximal end position a proximal end wall 154 of the trigger sleeve 150 abuts a sloped release surface 163 of each clamping arm 161. If the trigger sleeve 150 is further moved in its proximal end position the end wall 154 presses the clamping arms 161 radially outwards and thus compressing the arm springs 162 as shown in FIG. 13. This releases the clamping arms from the ledges 133 and thus the syringe unit 102 is released from the drive unit 103. That means the compressed syringe connector spring 148 can relax thereby moving by the syringe connector 147 the entire unlocked syringe unit 102 in distal direction. The user can then remove the syringe unit 102 out of the drive unit housing 180. The controller drives the motor to move the trigger sleeve 150 distally back in its initial position. The autoinjector 100 is now ready to be loaded with a new syringe unit 102.

[0189] FIG. 14 depicts a third embodiment with an autoinjector 200 according to the disclosure. In contrast to the second embodiment the drive unit 203 of the autoinjector 200 includes a slider 210 that can be shifted along the longitudinal axis by the user in order to move the syringe unit 202 out of the drive unit after an injection. Hence, in contrast to the above described last step (with respect to FIG. 13) the clamping arms are not deflected by a trigger sleeve movement but by a manual movement of the syringe unit by the user. All other features of the third embodiment are identical the above described second embodiment.

[0190] FIGS. 15 to 17 show a further embodiment of the disclosure. FIG. 15 shows a system according to the disclosure showing the injection device 301, an exchangeable cartridge unit 310 which may be inserted into the injection device 301, and smart phone 320. The NFC tag 311 on the cartridge unit housing 310a is depicted as a symbolic antenna. FIG. 16 shows the injection device 301 with the outer distal housing 301b removed, such that the near field communication module, i.e., the NFC receiver and transmitter 302 becomes visible-also depicted as symbolic antenna. The cartridge module 310 includes a prefilled syringe 312 (see FIG. 17). Within the syringe a piston 314 is disposed, which is movable between proximal position (as in FIG. 17) and a distal position where piston reaches the shoulder of the syringe body. As the piston is moved into distal direction, fluid medicament is ejected through the syringe cannula 313. Before use, the cannula is protected by a needle shield 315 which is removed by the needle shield removing unit 316 when the injection device 301 is to be used to deliver medicament.

[0191] Injection device 301 is a motor driven autoinjector. It includes an electronic controller, a drive assembly including the motor and the NFC receiver and transmitter. A person skilled in the art is aware of motor driven autoinjectors, e.g., US 20200164145 A1 is a known example. The injection device 301 includes an inner housing 301c and outer housings, namely the proximal outer housing 301a and distal outer housing. After insertion of the cartridge unit 310 is detected by the injection device, e.g., by sensors in communication with the controller, the controller activates the NFC receiver and transmitter 302 to read the information contained in the NFC Tag 311. The read information can be used by the injection device for the preparation of the medicament injection. The controller may further use the

read information to check the type of medicament and e.g., the shelf life. If the expiration date is reached the injection device 301 may emit a signal to the person who intends to use the device-therefore improving safety of the medicament use.

[0192] The NFC receiving and transmitting 302 device may be used to transmit data and further information to the smart phone 320 which includes an appropriate reading unit. In order to transmit data and further information from the injection device 301 to the smart phone 320 the controller of the injection device 301 sets the NFC receiver and transmitter 302 into the so-called card emulation mode and provide the data and further information the NFC receiver and transmitter 302. The smart phone 320 and the injection device 301 are then moved to close proximity or directly adjacent which allows the reading unit of the smart phone to extract the data and further information from the NFC receiver and transmitter 302. The smart phone 320 is configured with software to analyze and process the received data and further information. For example transmitted error messages may be displayed on the display 321 of the smart phone 320. Moreover, the smart phone 320 is configured to forward at least part of the read data and further information, notably in processed form, to further devices, e.g., distant servers. This may be done through a local area network (LAN) connection, particularly through wireless LAN (WLAN) or a cellular or mobile network.

[0193] While the disclosure is described in detail in the drawings and foregoing description, such description is to be considered illustrative or exemplary and not restrictive. Variations to the disclosed embodiments will be understood and effected by those skilled in the art and practising the claimed disclosure, from a study of the drawings, the disclosure, and the appended claims.

[0194] In the claims, the word “including” does not exclude other elements or steps, and the indefinite article “a” or “an” does not exclude a plurality. For example “an arm” does not exclude the fact that there may be two arms that functionally or structurally fulfill the purpose of “an arm”. The mere fact that certain elements or steps are recited in distinct claims shall not preclude the existence of further meaningful combinations of these elements or steps.

LIST OF DESIGNATIONS

1	autoinjector
2	syringe unit
3	drive unit
4	button
10	cap
11	deflectable tongues
12	holder
15	prefilled syringe
16	piston
17	rigid needle shield
18	injection needle
20	cover sleeve
21	first opening
22	second opening
23	indentation
24	stop surface
25	label
26	RFID tag
27	release surface
30	syringe holder
31	longitudinal rail
32	locking arms

-continued

LIST OF DESIGNATIONS	
33	ledge
34	bearing surface
40	plunger rod
41	threaded rod
45	cover sleeve connector
46	cover sleeve spring
47	syringe connector
48	syringe connector spring
49	flange
50	trigger sleeve
51	trigger sleeve connector
52	ledge
53	trigger sleeve spring
54	trigger arm
55	arm sloped contact surface
56	free end
60	gripper sleeve
61	clamping arms
62	clamp
64	gripper sleeve spring
65	counter release surface
80	housing
81	distal housing insert
82	contact surface
83	sleeve mechanism
84	wall
85	proximal housing insert
86	protrusion
91	electric motor
92	pinion
93	gear
94	battery
100	autoinjector
102	syringe unit
103	drive unit
110	cap
120	cover sleeve
130	syringe holder
132	locking arms
133	ledges
135	cap holding tongues
136	protrusion
137	ledge
140	plunger rod
141	threaded rod
145	cover sleeve connector
146	cover sleeve spring
147	syringe connector
148	syringe connector spring
150	trigger sleeve
151	control cam
152	first contact surface
153	third contact surface
154	end wall
155	cap cam
156	cap distal surface
157	cap prox. surface
158	trigger sleeve connector
159	trigger sleeve spring
160	intermediate sleeve
161	clamping arms
162	spring
163	release surface
170	lever
171	counter cam
172	second contact surface
173	fourth contact surface
175	counter cap cam
176	counter distal surface
177	counter prox. surface
178	actuation nose
179	cap actuation nose
180	housing
181	distal housing insert

-continued

LIST OF DESIGNATIONS	
183	sleeve mechanism
191	motor
200	autoinjector
202	syringe unit
203	drive unit
210	slider
301	autoinjector
301a	proximal outer housing
301b	distal outer housing
301c	inner housing
302	NFC receiver and transmitter
303	cartridge unit cavity
310	cartridge unit
310a	cartridge unit housing
311	NFC tag
312	prefilled syringe
313	syringe cannula
314	syringe piston
315	needle shield
316	needle shield removing unit
320	smart phone
321	display

What is claimed is:

1. A reusable drive unit of a drug delivery device for dispensing a liquid drug from a reservoir through a needle of a reservoir unit, wherein the drive unit is configured to be connected to the reservoir unit and comprises:

- a housing defining a longitudinal axis;
- an electronic module arranged in an interior of the housing, the electronic module comprising a controller;
- a connector arranged in the housing interior, the connector biased in a distal direction by a spring and configured to be axially movable;
- a detection element connected to the connector, wherein the connector and detection element are axially movable relative to the housing during attachment of a reservoir unit to the drive unit; and
- a position sensor arranged in the housing interior and communicatively coupled to the controller and configured to sense an axial movement of the detection element,

wherein when the connector and the detection element are axially moved, the sensor is configured to transmit a signal to the controller corresponding to a sensed axial movement of the detection element such that the controller detects a presence of a reservoir unit in the drive unit.

2. The drive unit of claim 1, wherein the detected presence of the reservoir unit in the drive unit corresponds to the reservoir unit being in a locked state with the drive unit.

3. The drive unit of claim 2, wherein the drive unit is configured to establish the locked state with the drive unit by a releasable connection in a snap-fit, a bayonet connection or a thread.

4. The drive unit of claim 3, wherein the connector comprises a distally directed surface configured to contact a proximally directed surface of the reservoir unit to cause the axial movement of the detection element.

5. The drive unit of claim 1, wherein the connector comprises a distally directed surface configured to contact a proximally directed surface of the reservoir unit to cause the axial movement of the detection element.

6. The drive unit of claim 1, wherein the drive unit comprises a plunger rod coaxially arranged within the connector and the spring, the plunger rod configured to be moved through an opening defined in the connector in the distal direction in a dispensing movement to cause the liquid drug to be dispensed from the reservoir of the reservoir unit.

7. The drive unit of claim 6, wherein the plunger rod is non-rotatable relative to the housing during the dispensing movement.

8. The drive unit of claim 7, wherein the plunger rod is in a threaded engagement with an automatic drive configured cause the plunger rod to be moved in the dispensing movement.

9. The drive unit of claim 1, wherein the position sensor is configured as a magnetic sensor.

10. A drug delivery device comprising a reusable drive unit and a reservoir unit, comprising:

- a drive unit housing defining a longitudinal axis;
- an electronic module arranged in an interior of the housing, the electronic module comprising a controller;
- a connector arranged in the housing interior, the connector biased in a distal direction by a spring and configured to be axially movable;

a detection element connected to the connector, wherein the connector and detection element are axially movable relative to the housing during attachment of the reservoir unit to the drive unit; and

a position sensor arranged in the housing interior and communicatively coupled to the controller and configured to sense an axial movement of the detection element,

wherein the connector comprises a distally directed surface configured to contact a proximally directed surface of the reservoir unit to cause the axial movement of the detection element, and

wherein when the detection element is axially moved, the sensor is configured to transmit a signal to the controller corresponding to a sensed axial movement of the detection element such that the controller detects a presence of the reservoir unit in the drive unit.

11. The drug delivery device of claim 10, wherein the detected presence of the reservoir unit in the drive unit corresponds to the reservoir unit being in a locked state with the drive unit.

12. The drug delivery device of claim 11, wherein the drive unit is configured to establish the locked state with the drive unit by a releasable connection in a snap-fit, a bayonet connection or a thread.

13. The drug delivery device of claim 12, wherein the position sensor is configured as a magnetic sensor.

14. The drug delivery device of claim 10, wherein the drive unit comprises a plunger rod coaxially arranged within the connector and the spring, the plunger rod configured to be moved through an opening defined in the connector in the distal direction in a dispensing movement to cause a liquid drug to be dispensed from a reservoir of the reservoir unit.

15. The drug delivery device of claim 14, wherein the plunger rod is non-rotatable relative to the housing during the dispensing movement, and wherein the plunger rod is in a threaded engagement with an automatic drive configured cause the plunger rod to be moved in the dispensing movement.

16. The drug delivery device of claim 10, wherein the reservoir unit comprises a needle cover sleeve with a retracted position and a needle covering position.

17. The drug delivery device of claim 16, wherein the position sensor is configured to transmit a signal to the controller corresponding to a sensed movement of the needle cover sleeve.

18. The drug delivery device of claim 17, wherein a movement of the needle cover sleeve is guided by the reservoir unit.

19. The drug delivery device of claim 10, wherein the reservoir unit comprises a tag and the drive unit comprises a tag reader for reading the tag of reservoir unit.

20. The drug delivery device of claim 19, wherein the tag is configured as an NPC tag or a RFID tag.

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